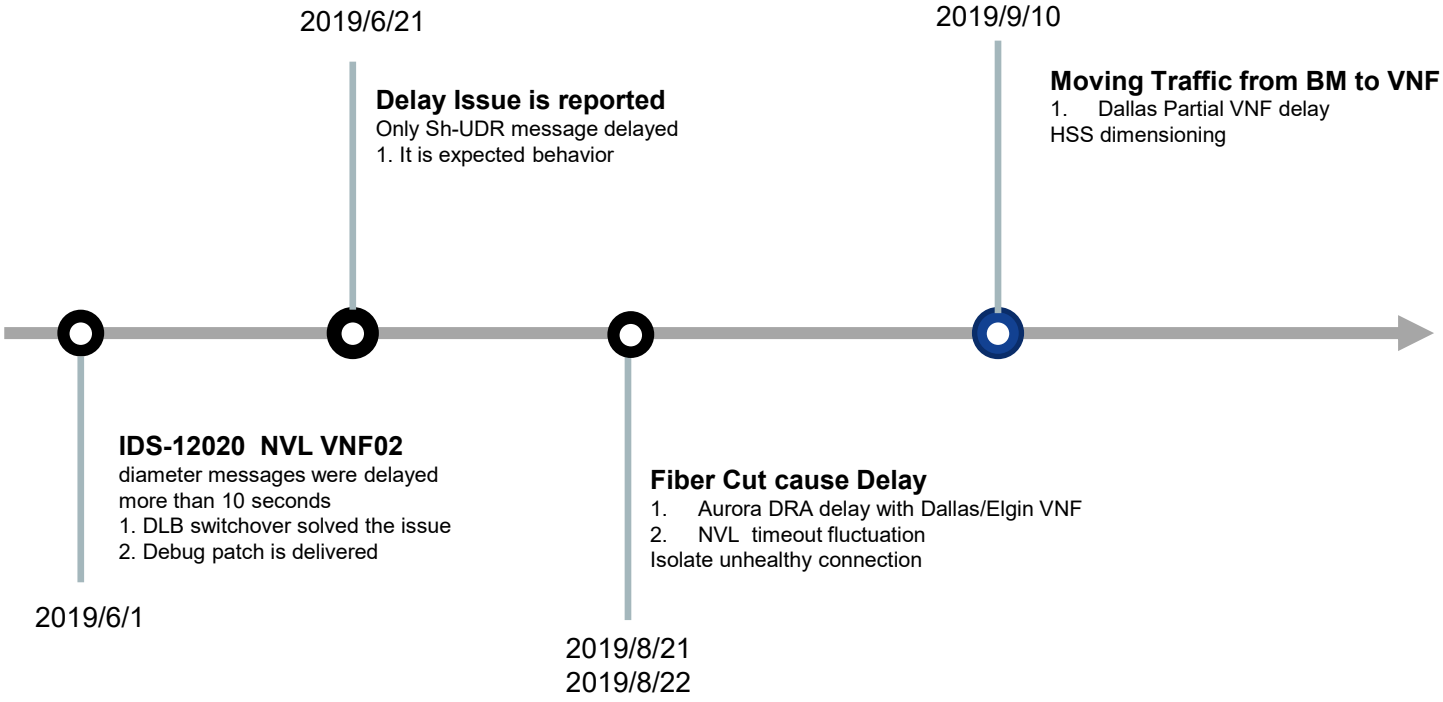
TMO Delay Issue

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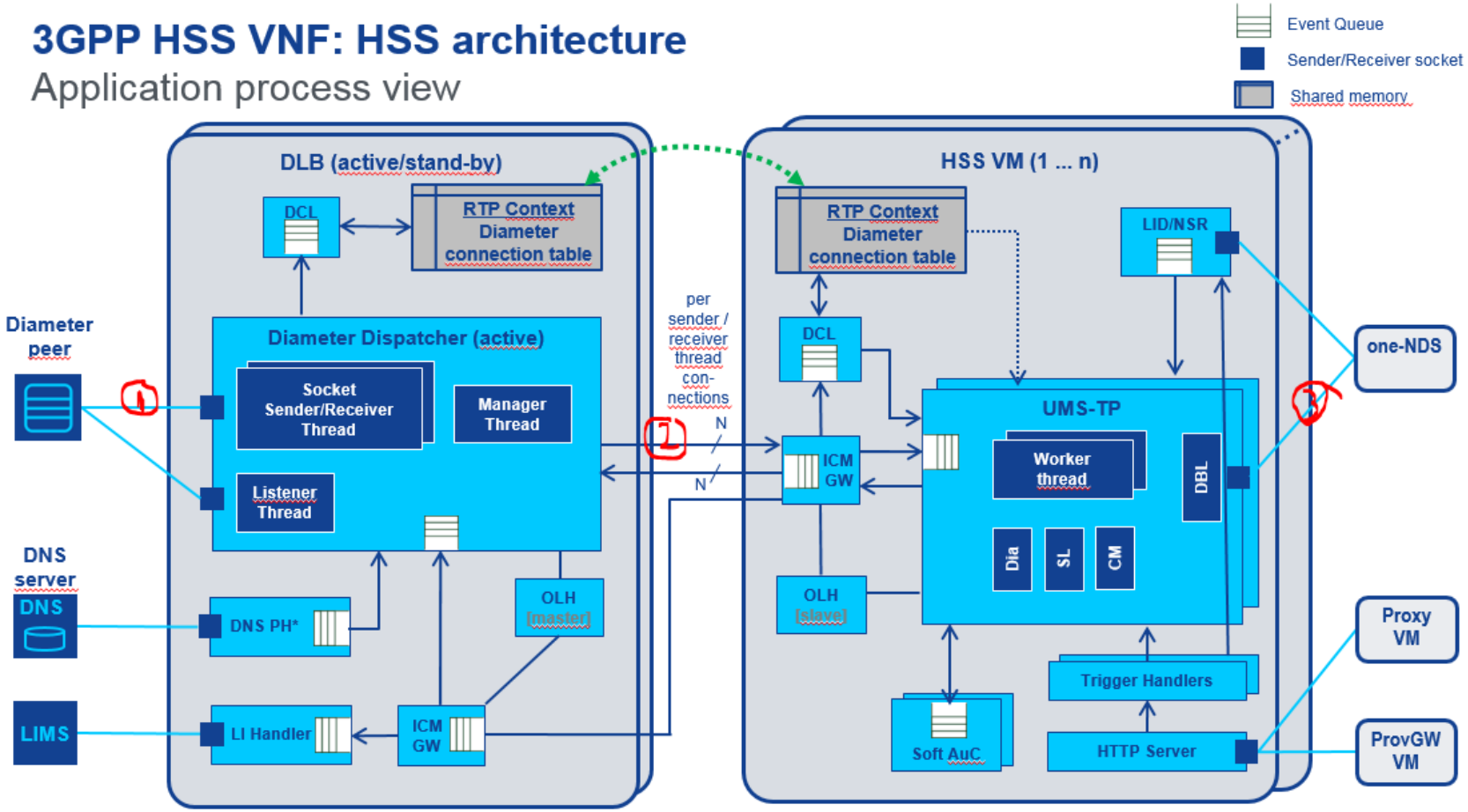
TMO Delay Issues Timeline



NOKIA

3GPP HSS VNF: HSS architecture

Application process view



Tips

1. Single process handling multiple task: Not use blocking API

2. when building debug patch use –release option :
   gm_git DiaDispatcher -cld -release -gf -j12

3. When doing feature, pay more attention to Log/trace
4. Do not exclude investigation direction unless you have concrete evidence
5. Decease manual operations in debug patch

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First Outage Summary

Description:

A fibre cut triggered service impact in Elgin and Dallas sites, particularly with the Nokia HSS and TMO DRA.

High delays in messages arriving at the Aurora DRA, resulting in messages arriving after the timeout at the DRA had expired.

Elgin site returned to normal function

Dallas site continued to experience message delays seen at the DRA from the HSS, traffic was offloaded from the HSS to restore normal function across the entire platform.

There is an ongoing Nashville EME for timeouts towards DRA from HSS

Cause:

- Triggering event, a fibre cut
- High message delays seen at the DRA from HSS, cause yet unknown

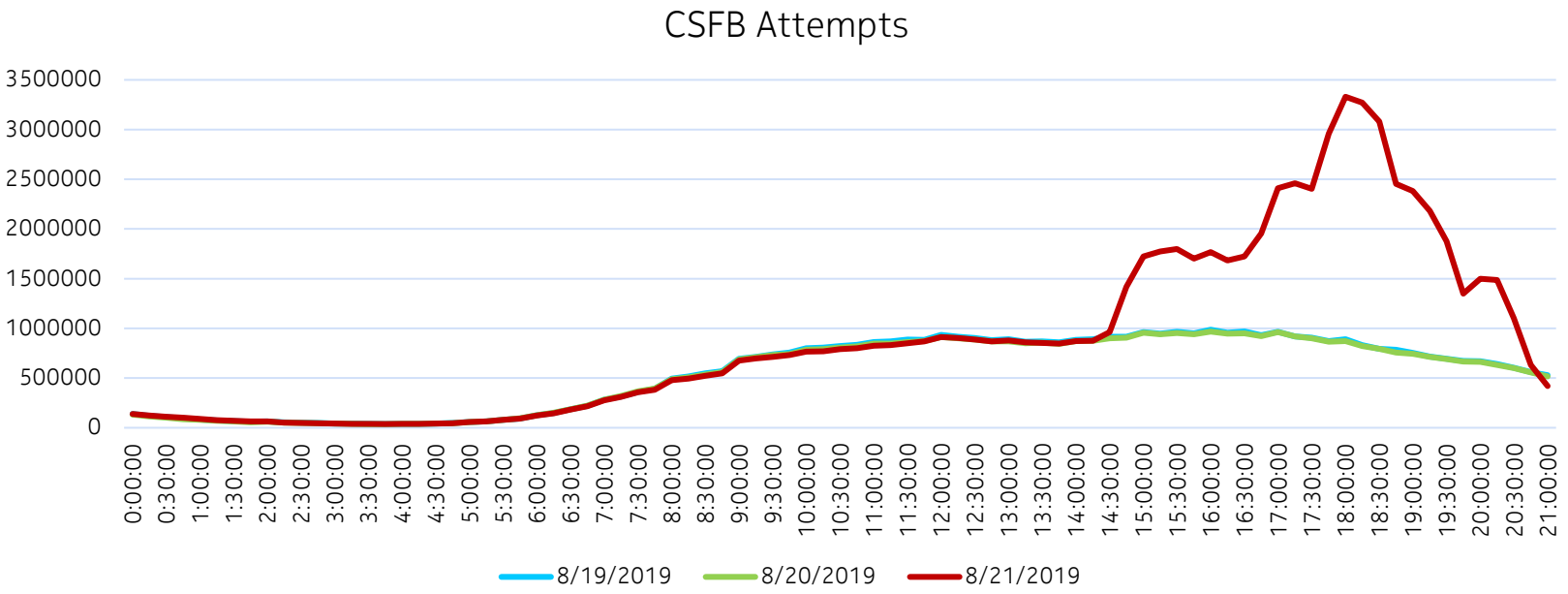
Recovery actions:

- Aurora DRA shutdown link towards Elgin HSS
- All MME isolated from Aurora DRA
- Traffic was offloaded from Dallas HSS

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Frist Outage Impact --CSFB Attempts

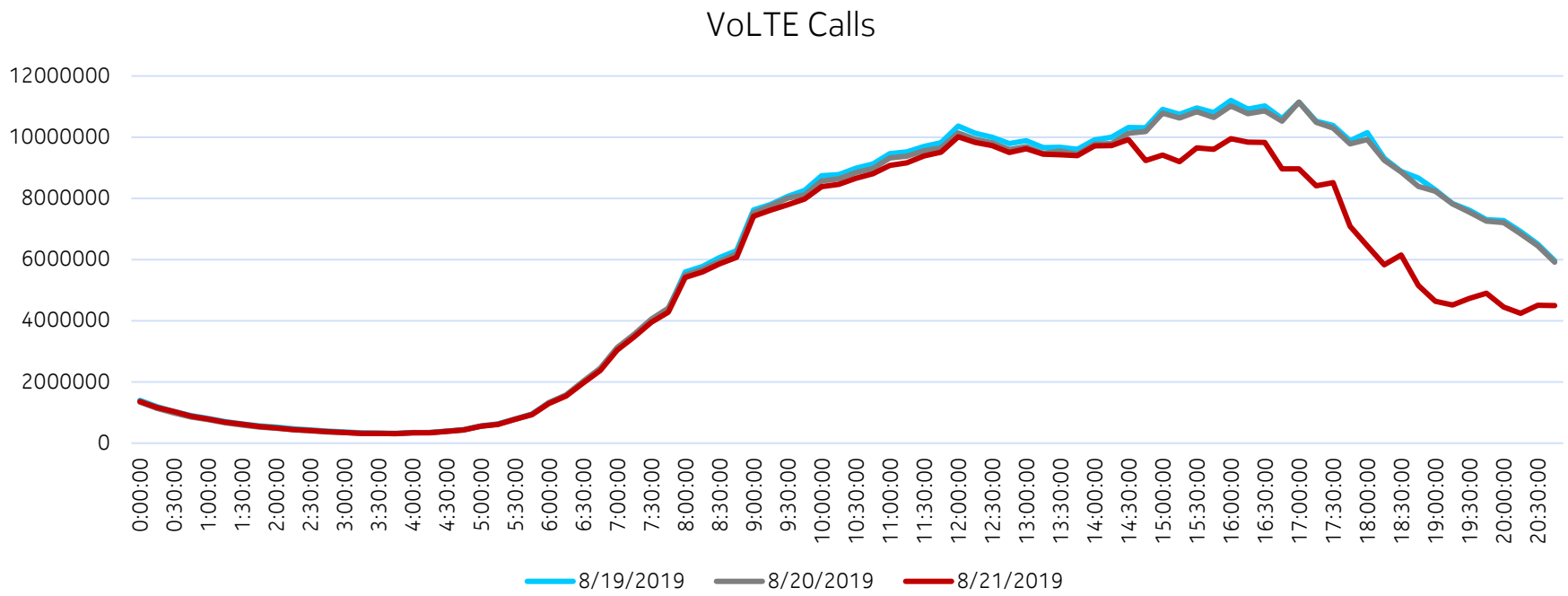


8.1 Million (31%) CSFB Call Increase

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Frist Outage Impact --VoLTE Calls

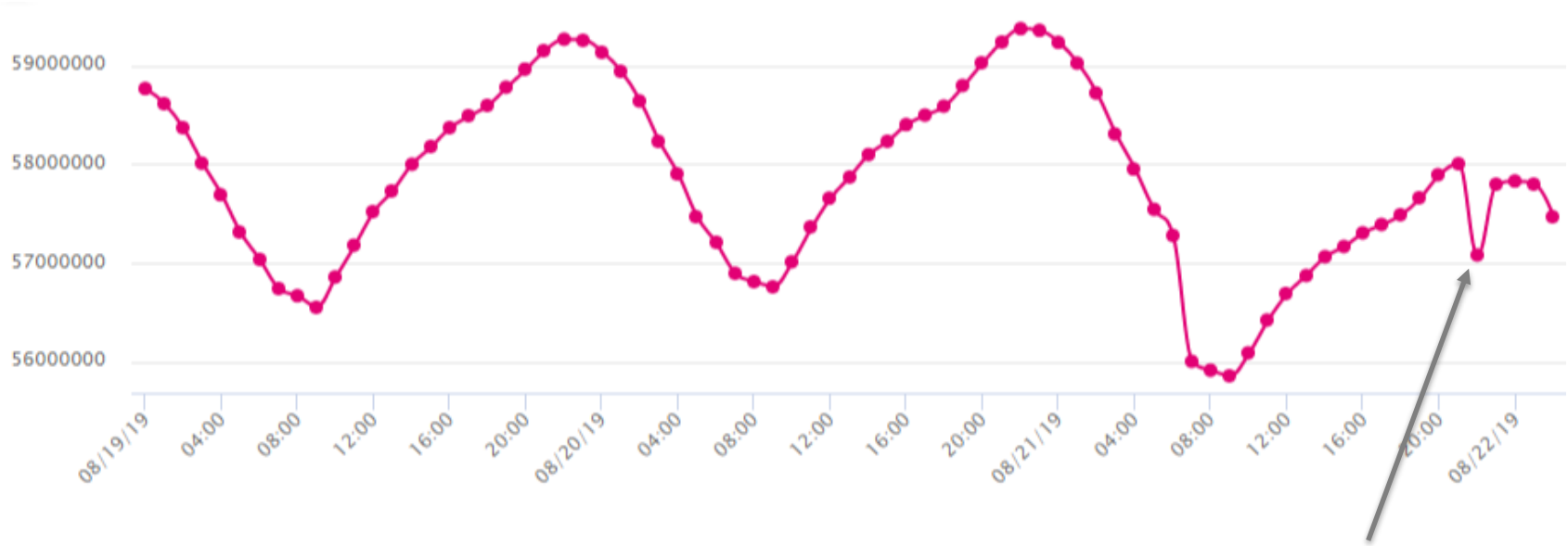


54.4 Million (25%) VoLTE Call Decrease. VoLTE call count continues to trend lower

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Frist Outage Impact --LTE MME session



2.289 million subscriber reduction at peak

T-Mobile Confidential

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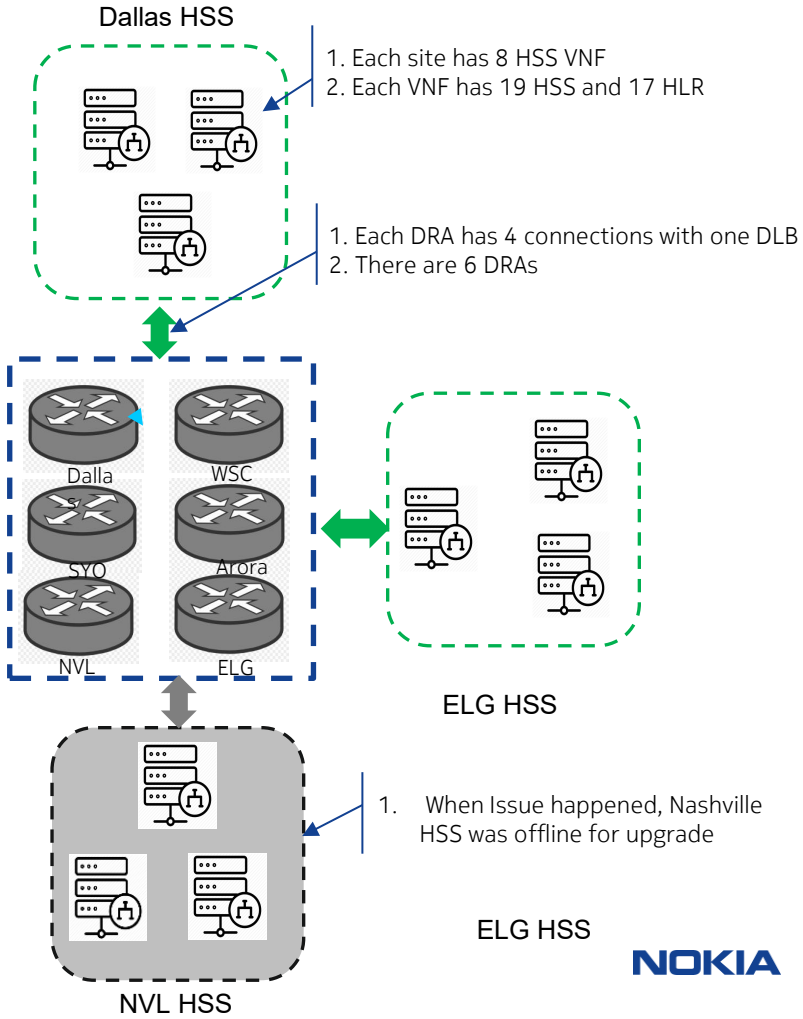
Frist Outage Impact --IMS Call

- From TAS CDRs we observed 480 Cause Codes spiked when this outage started. Count: 28,175,810
- From MSS CDRs we observed CS to IMS call failures. Count: 32,301,978
- From SBI CDRs we observed inbound MGCF Bypass call failures. Count: 38,846,969
- **Total call failures: 99,324,757**

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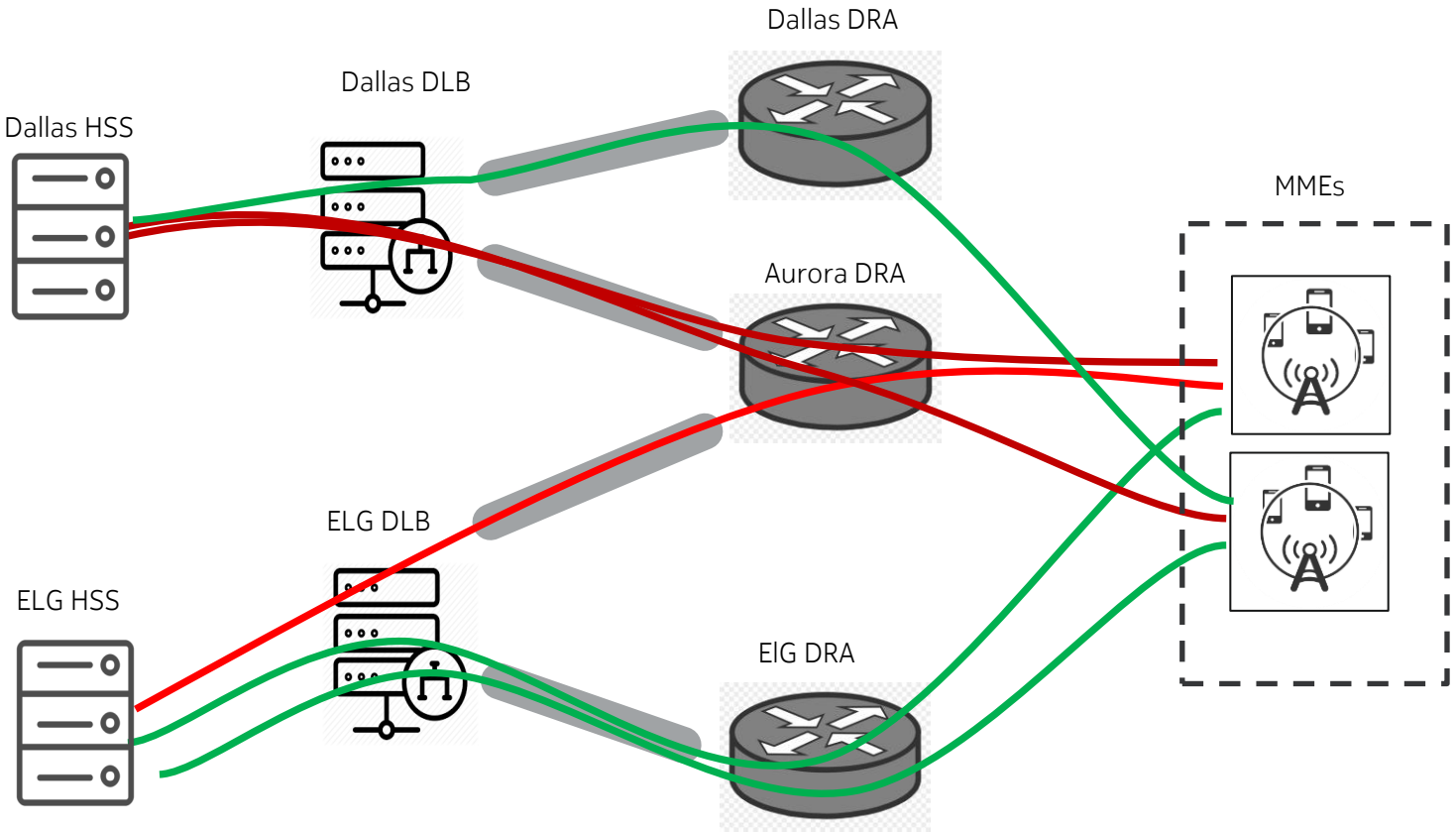
TMO First Outage Timeline

	Events
2:30pm PT	DRA timeout started – from Aurora DRA to Dallas & Elgin HSS
2:40pm PT	Degradation began
2:45pm PT	Spike in Diameter unable to deliver errors, Dallas DRA sees timeout to Dallas HSS, but not Elgin HSS Elgin DRA sees timeout to Dallas HSS, and also to Elgin HSS
3:27pm PT	ORCs bundle were brought down manually(Aurora DRA brought down)
3:45pm PT	Time out to Elgin HSS recovered
4:11pm PT	Aurora DRA shutdown link towards Elgin HSS
4:29pm PT	Aurora DRA totally offline
4:36pm PT	All MME isolated from Aurora DRA
5:30pm PT	Time out to Dallas stopped (traffic was offloaded)



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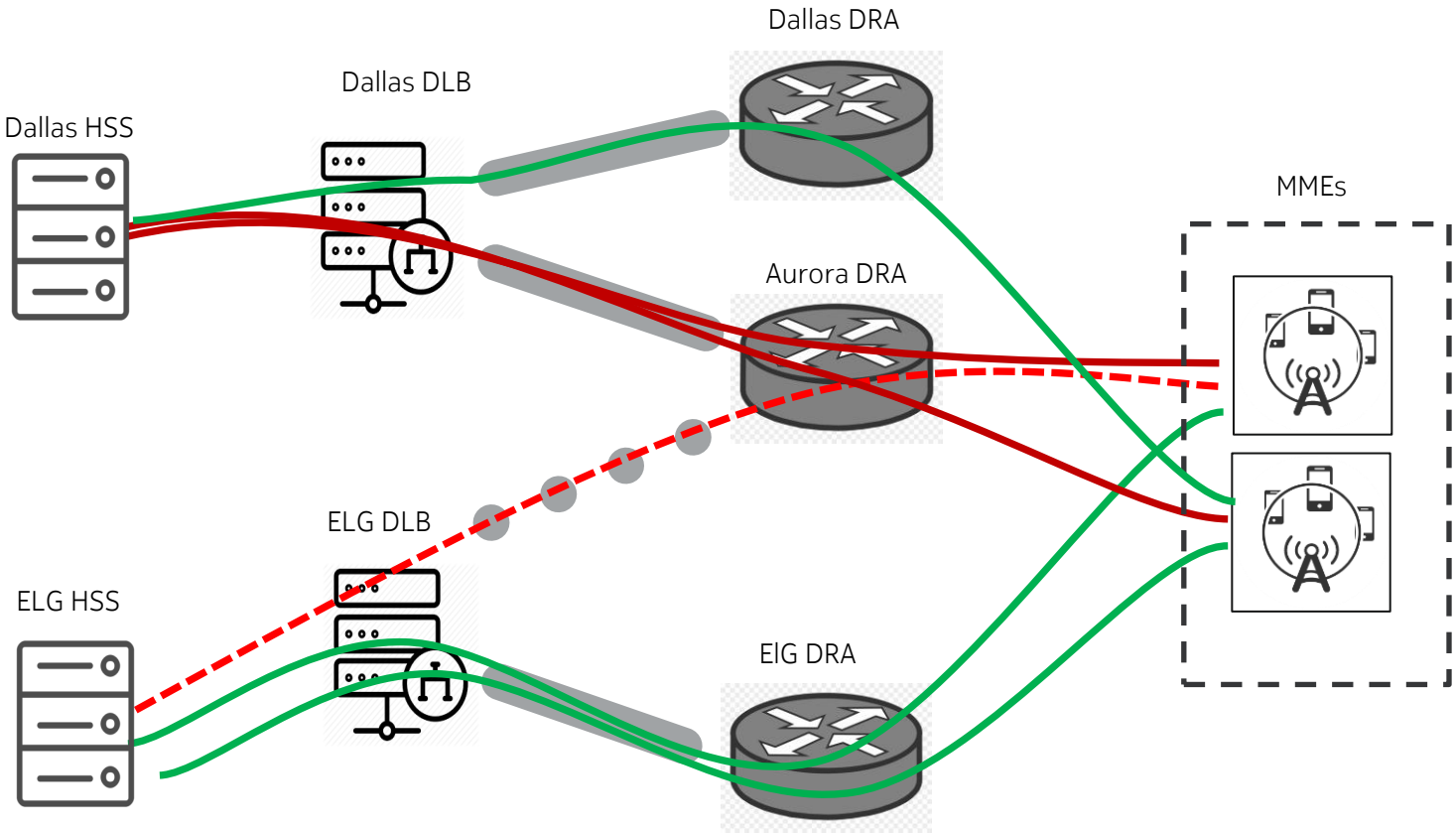
Fiber Cut – DRA Aurora impact



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Elgin – DRA ARO – connection brought down

2019/8/21 17:56 Cut down connection with ELG DLB
Elgin HSS start to recovery

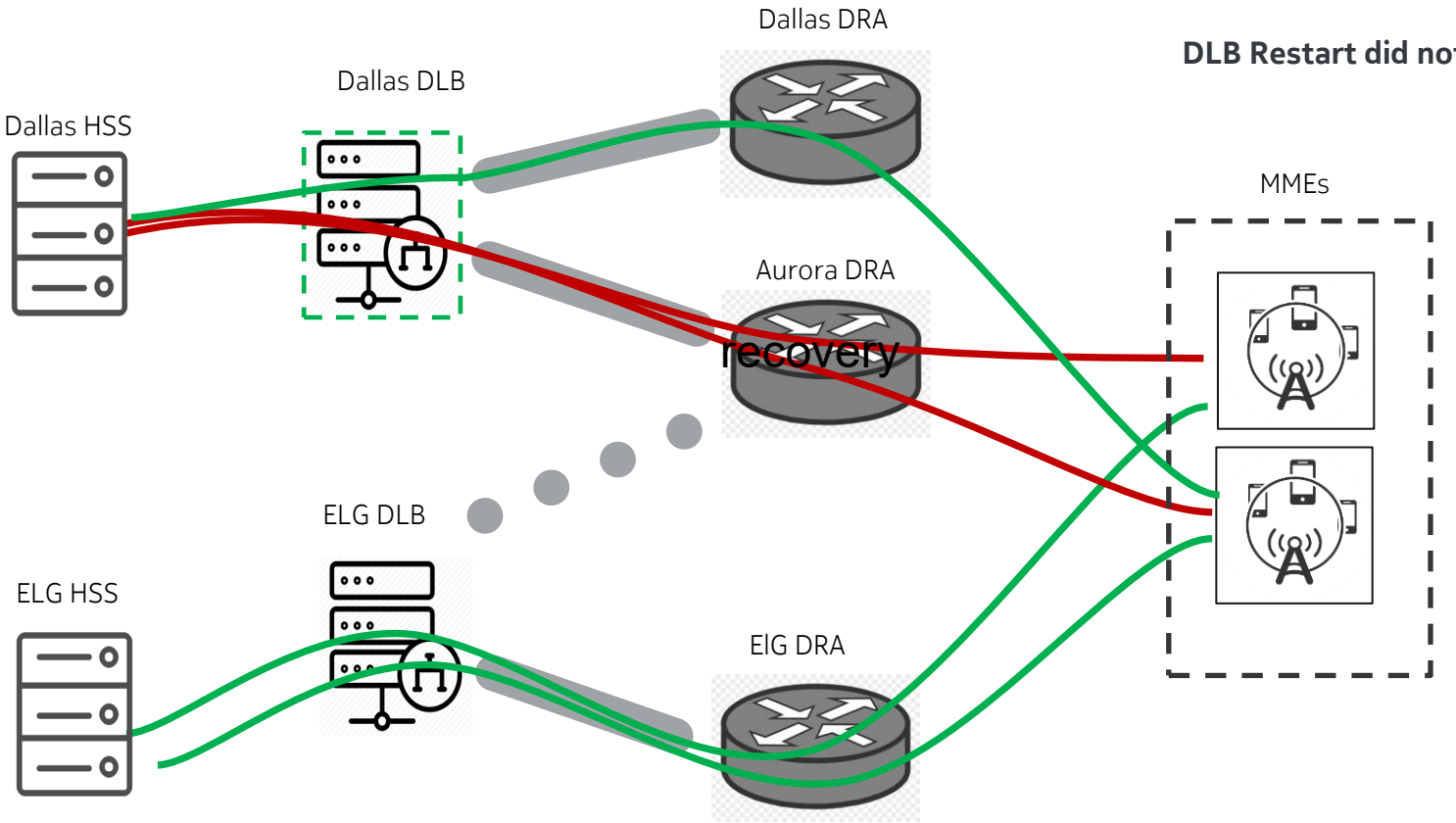


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Dallas HSS – Restart DLBs in 5 VNF

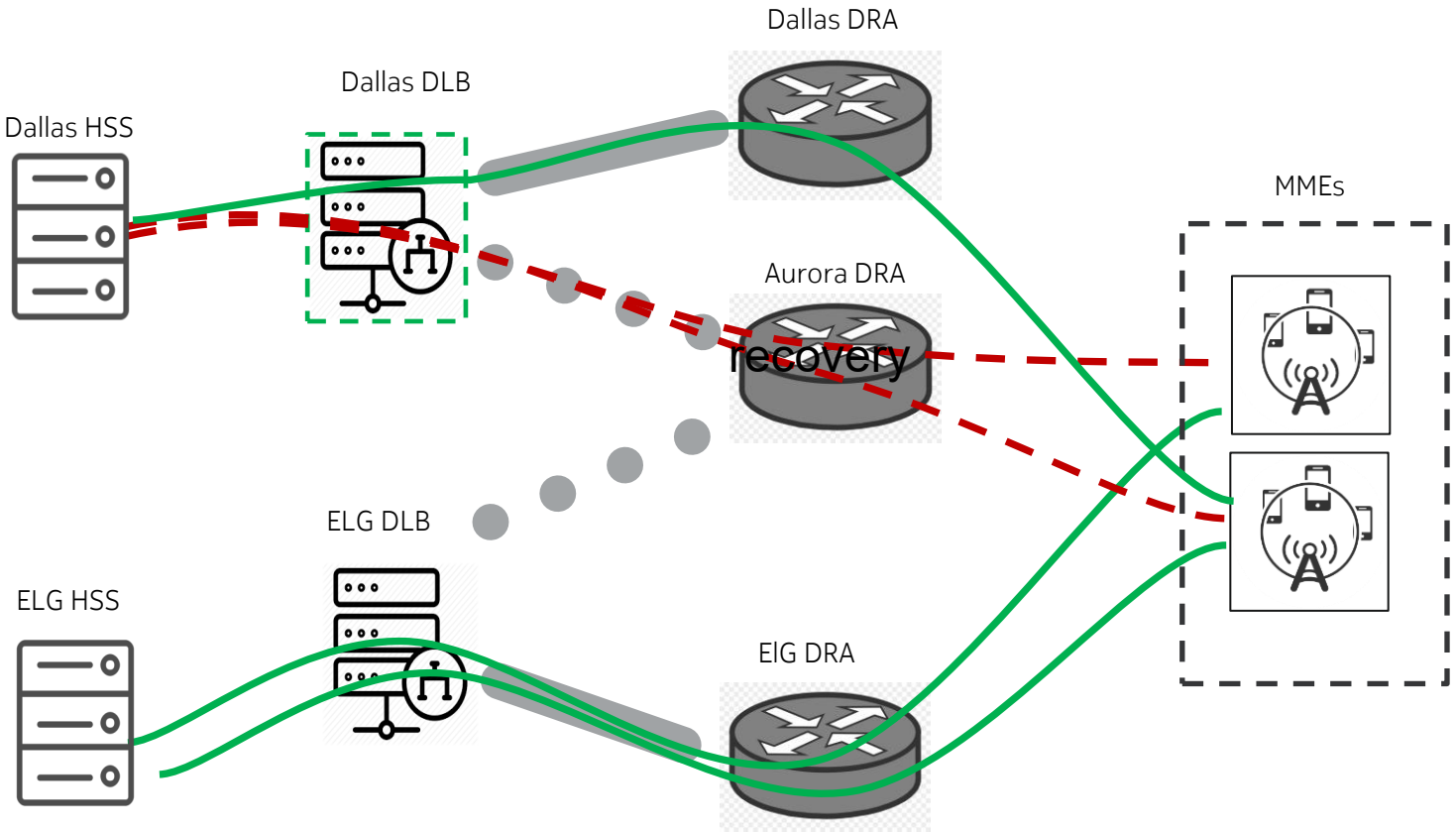
DRA – ARO still connected to HSS/DLB-DAL

DLB Restart did not produce results



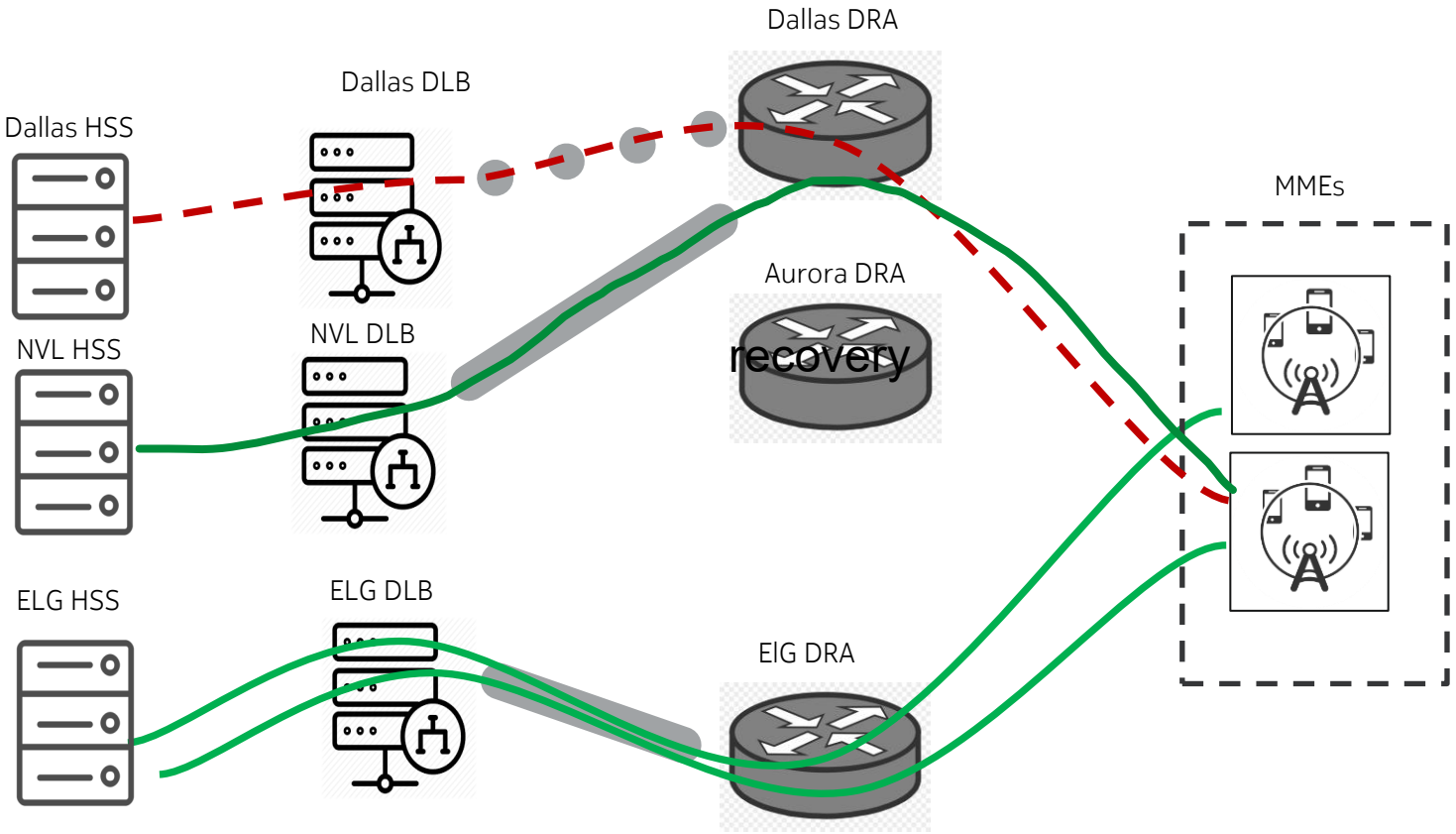
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Dallas HSS isolation with ARO DRA



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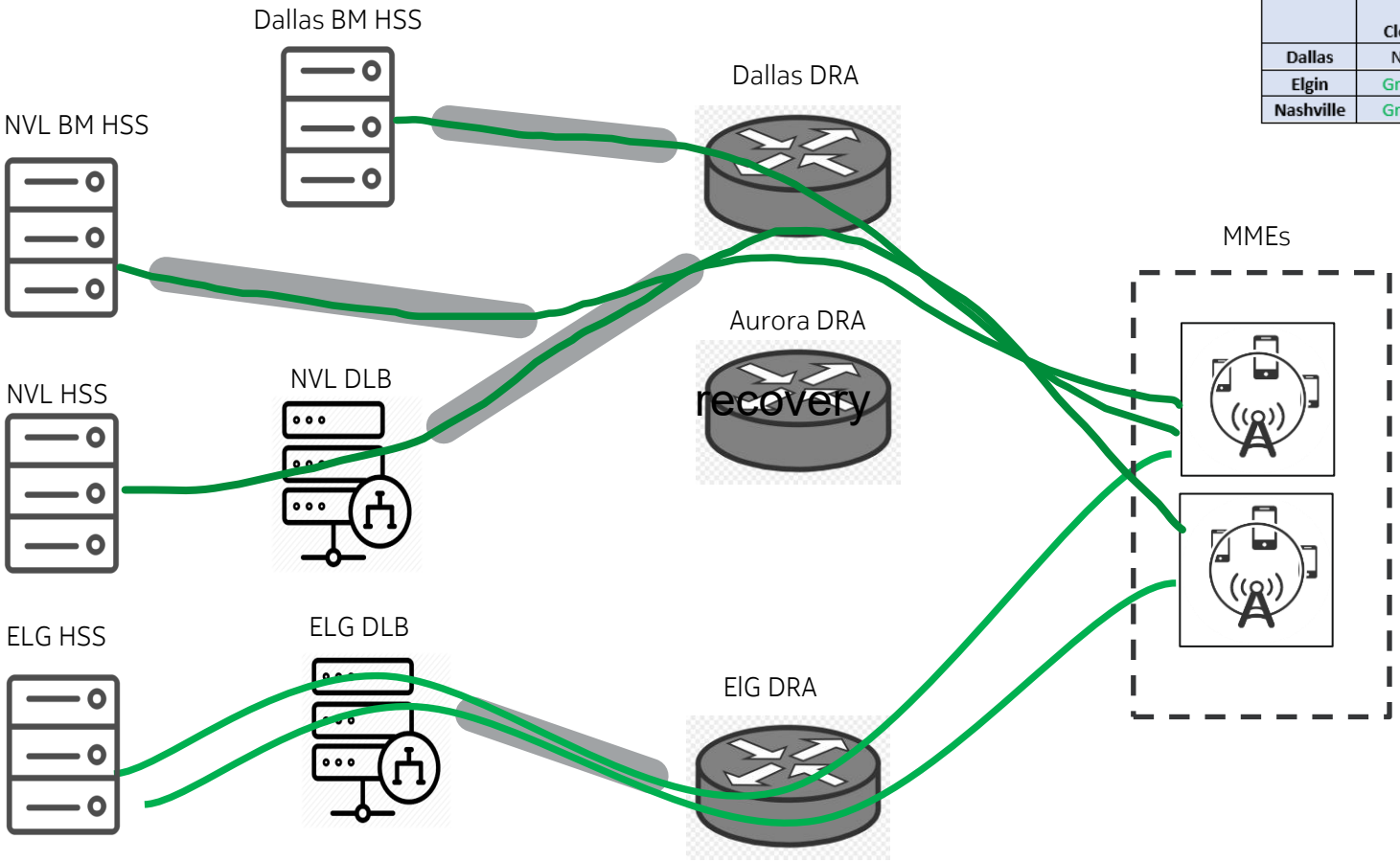
NVL HSS online and move traffic to NVL



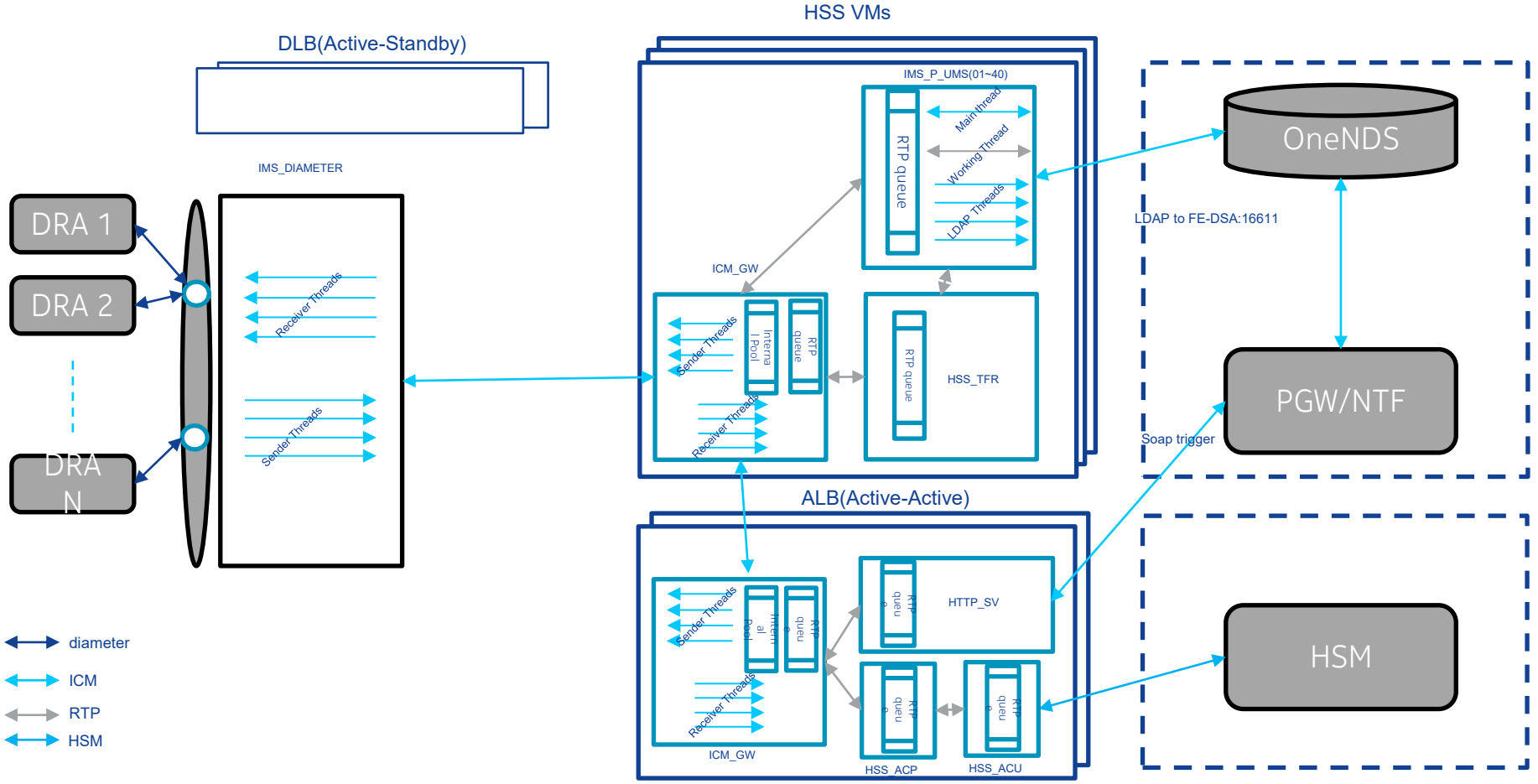
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NVL delay move traffic to Dallas BM HSS and NVL BM HSS

	Cloud	Bare Metal
Dallas	N/A	Green
Elgin	Green	N/A
Nashville	Green	Green



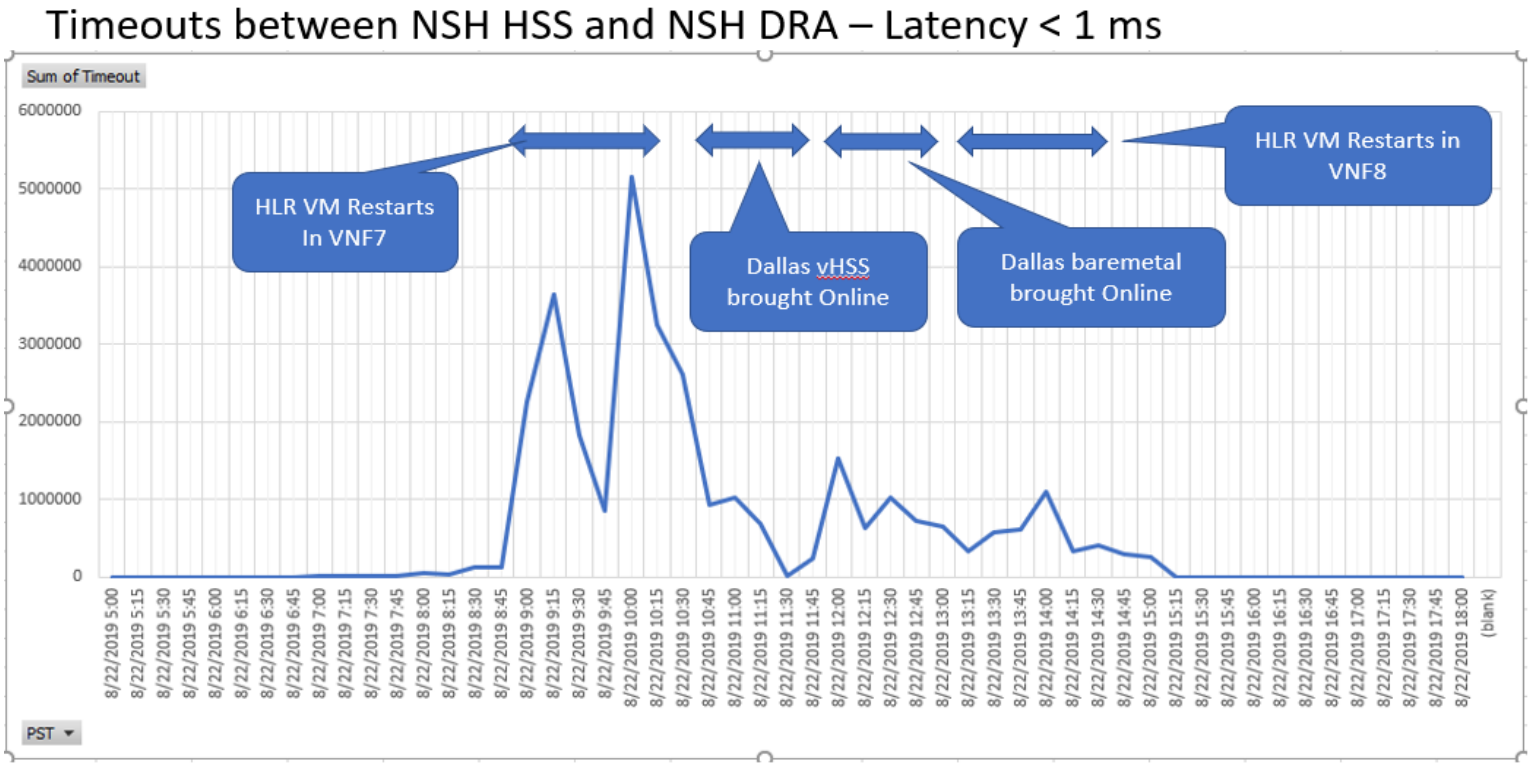
NOKIA



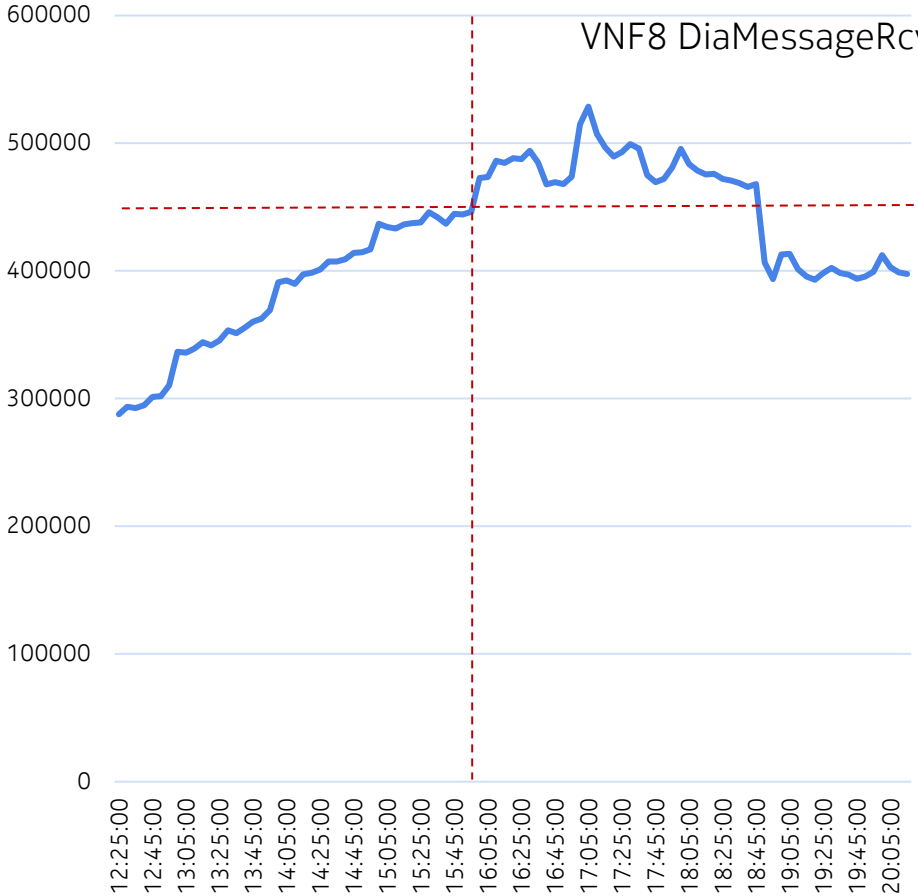
Code analysis and suggest about how to reproduce the problem

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NVL HSS site Timeout



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threshold

Conclusion：NVL issue can be explained, whenever TPS reach a threshold, HSS will delay message sending

Nokia Internal

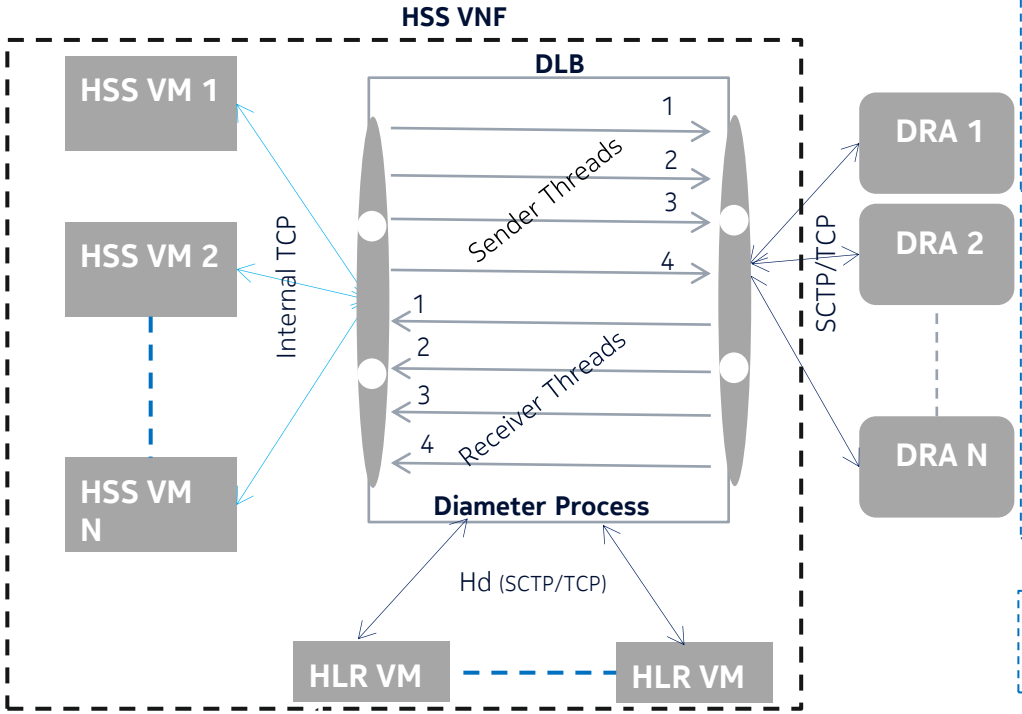
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HSS VNF Tunings

Socket poll time tuning

TCP Socket comm ⟷ Sender Threads → Socket and Buffer

SCTP Socket comm ⟷ Receiver Threads ←



Current State :

1. **DLB will have 24 SCTP Diameter connections from 6 DRAs, 4 SCTP connections from each DRA to every HSS.**
2. **When SCTP socket send buffer is full, DLB will try sending/writing packet with timed wait of 1000msec**

Next Step :suggest to do PT test and reproduce the problem with high WAN delay set on DRA.

Commands to add latency to SCTP paths:

sudo tc qdisc add dev bond0.5 root netem delay 50ms

sudo tc qdisc add dev bond0.6 root netem delay 50ms

Commands to remove latency to SCTP paths:

sudo tc qdisc del dev bond0.5 root netem delay 50ms

sudo tc qdisc del dev bond0.6 root netem delay 50ms

Result: TMO lab reproduce this issue ,325ms delay observed on Good DRA

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Solution to isolate unhealthy connections

1. Polling socket with timeout (1000ms → 100ms), if Timout , drop message



2. Check if EAGAIN counter > 10 within 5 seconds,
disconnects the connection and raise alarm

Solution :

1. **When SCTP socket send buffer is full, DLB will try sending/writing packet with timed wait of 100msec**
2. **If fails to write even with 100msec, it will discard the packet and count it as failed attempt**
3. **10 such attempts will be counted in a window of 5sec and SCTP association will be teared down**
4. **This timed wait will force other packets ready to be sent out to go in to wait state adding to delay**
5. **If more such attempts fail, more is the latency experienced by other packets ready to be sent out**

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Second Outage Summary

Description:

TMO moved Traffic from Dallas BM HSS to Dallas HSS VNF

Start from busy hour in the morning ,high delays in messages overserved only to Dallas HSS VNF
Especially to VNF 4, 5,7, 8. VNF4 is the worst

Customer expect RCA for delay issue and why only VNF 4, 5, 7, 8 have this issue.

Cause:

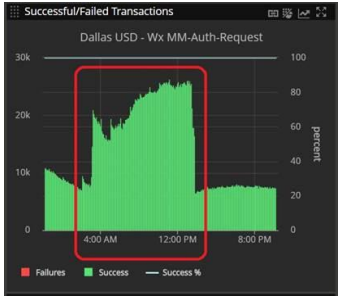
- Move Traffic from Dalla BM HSS to Dallas HSS VNF
- High message delays seen at the DRA from HSS

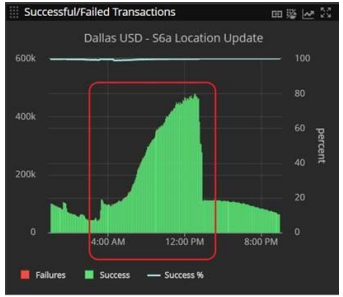
Recovery actions:

- Shutdown traffic from WSC DRA to VNF HSS
- Traffic was moved back to Dallas BM HSS

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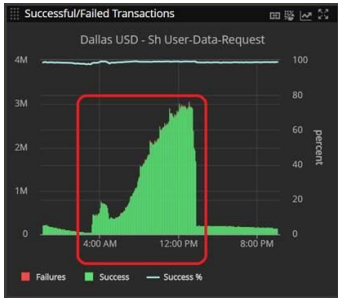
Second Outage Impact

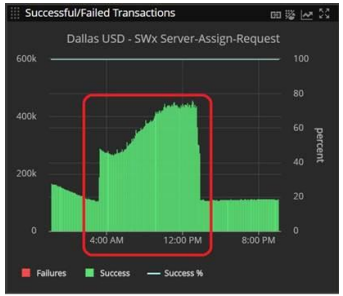


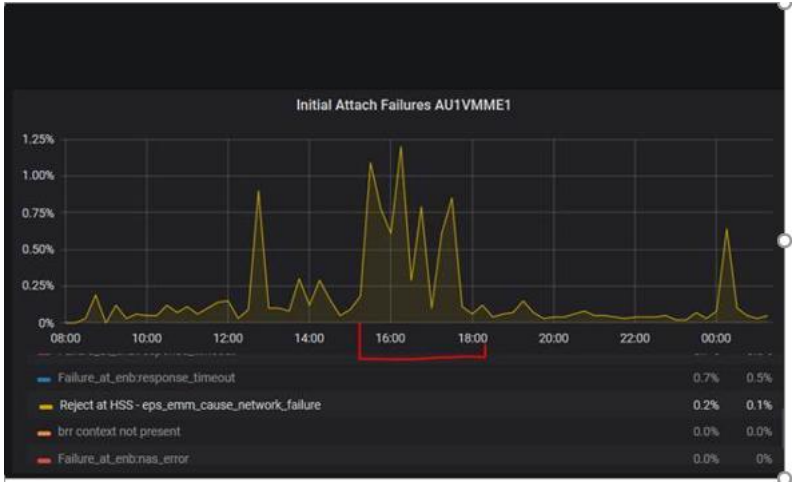








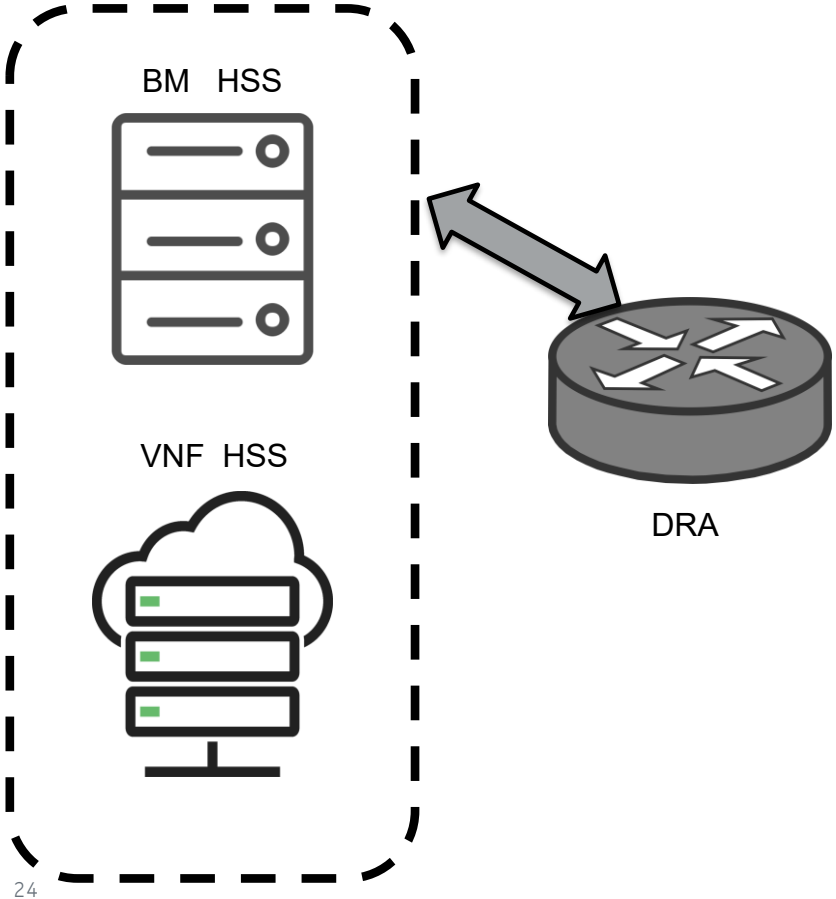




1. Dallas HSS has KPI degradation (dropped to 80%)
2. About 1% call failure for TMO

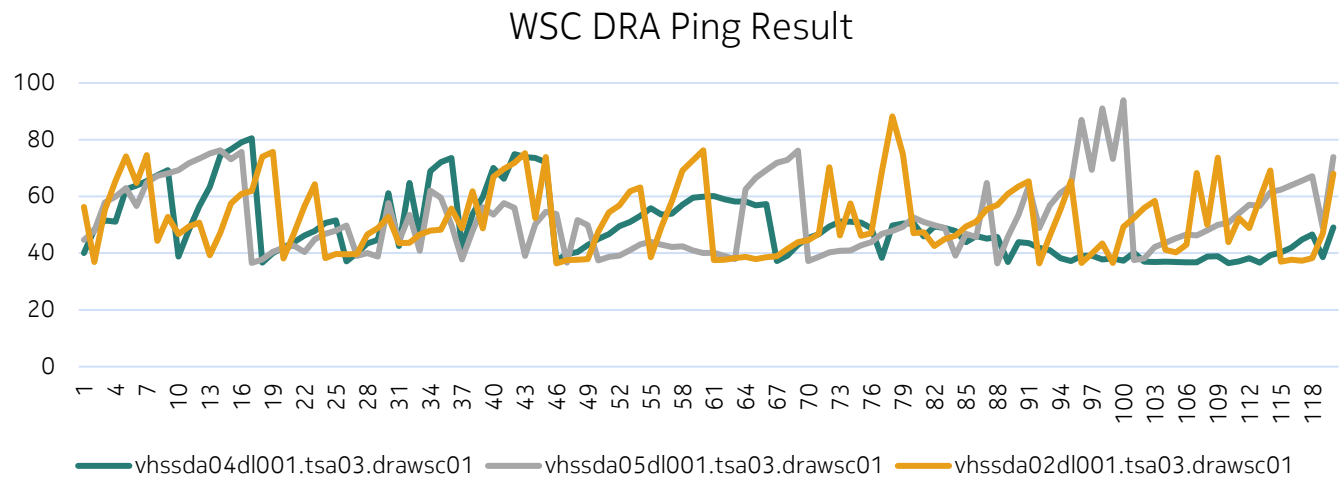
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Second Outage Timeline



- 2019/9/10 03:00 Move Traffic from BM to VNF
- 2019/9/10 06:56 VNF05 disconnected WSC DRA
- 2019/9/10 07:34 VNF04 disconnected WSC DRA
- 2019/9/10 07:45 VNF08 disconnected WSC DRA
- 2019/9/10 08:53 VNF07 disconnected WSC DRA
- 2019/9/10 12:27 Shutdown WSC DRA to VNF HSS
- 2019/9/10 12:44 Move Traffic from VNF to BM
- 2019/9/10 13:40 No timeout error, EME

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	VNF04dlb01(ms)	VNF05dlb01(ms)	VNF02dlb01(ms)
tsa01-drawsc01(10.177.70.134)	68.02966102	53.6059322	50.59661017
tsa03.drawsc01(10.177.70.136)	49.65916667	52.70083333	52.13583333
mp4.daldra01(10.160.113.12)	21.221025	16.07346667	19.803475

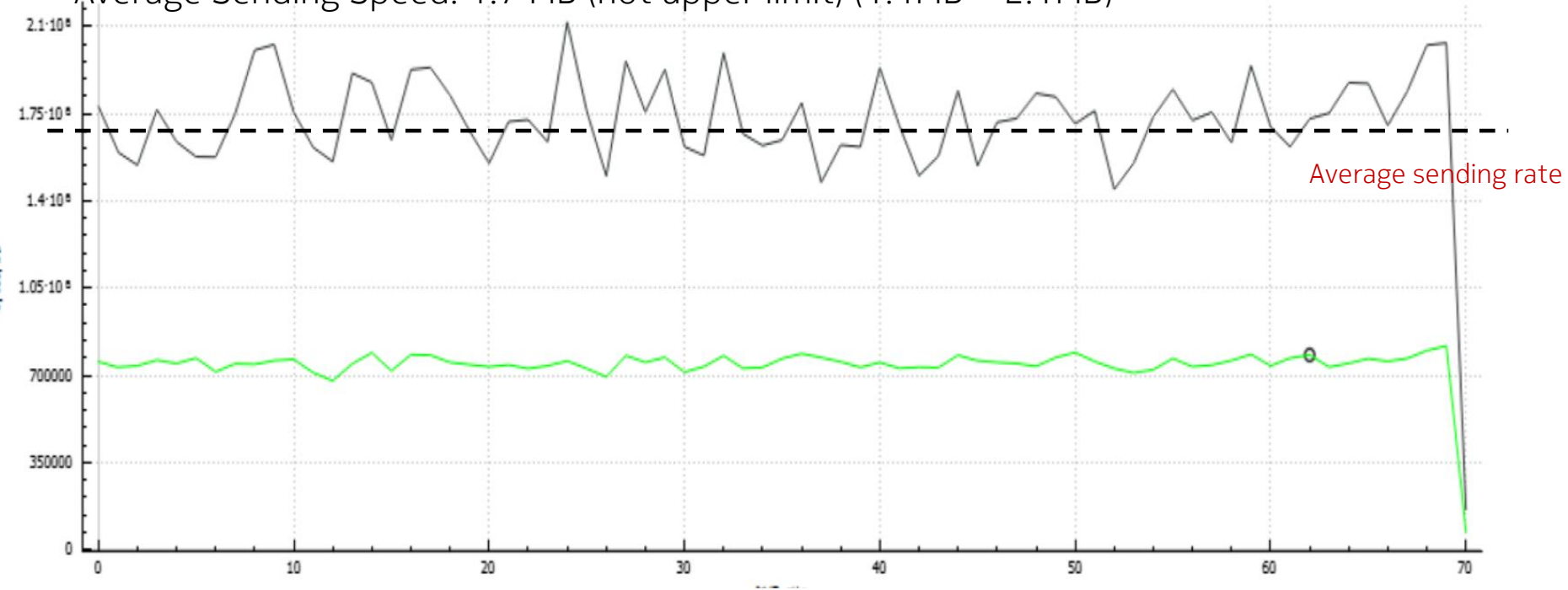
1. latency to WSC DRA is high specially for VNF4 (avg 68ms), which matches with high disconnection
2. From “10-09-2019 07:34:05” to “10-09-2019 12:24:50”, HSS intermittently disconnect the Diameter connection to WSC connection 89 times. From the netstat, the traffic are from both WSC and Dallas DRA. The send-Q of the connection to WSC DRA (>22K) is much higher than to Dallas DRA(1~3K). This is due to the WAN delay between Dallas HSS and WSC DRA (~60 ms)is higher than the one between Dallas HSS and Dallas DRA(16~30 ms)

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Healthy use case: VNF4 to Dallas DRA

Dallas DLB 04 ← → mp4.daldra01.dra.epc.mnc260.mcc310.3gppnetwork.org

Average Sending Speed: 1.7 MB (not upper limit) (1.4MB ~ 2.1MB)

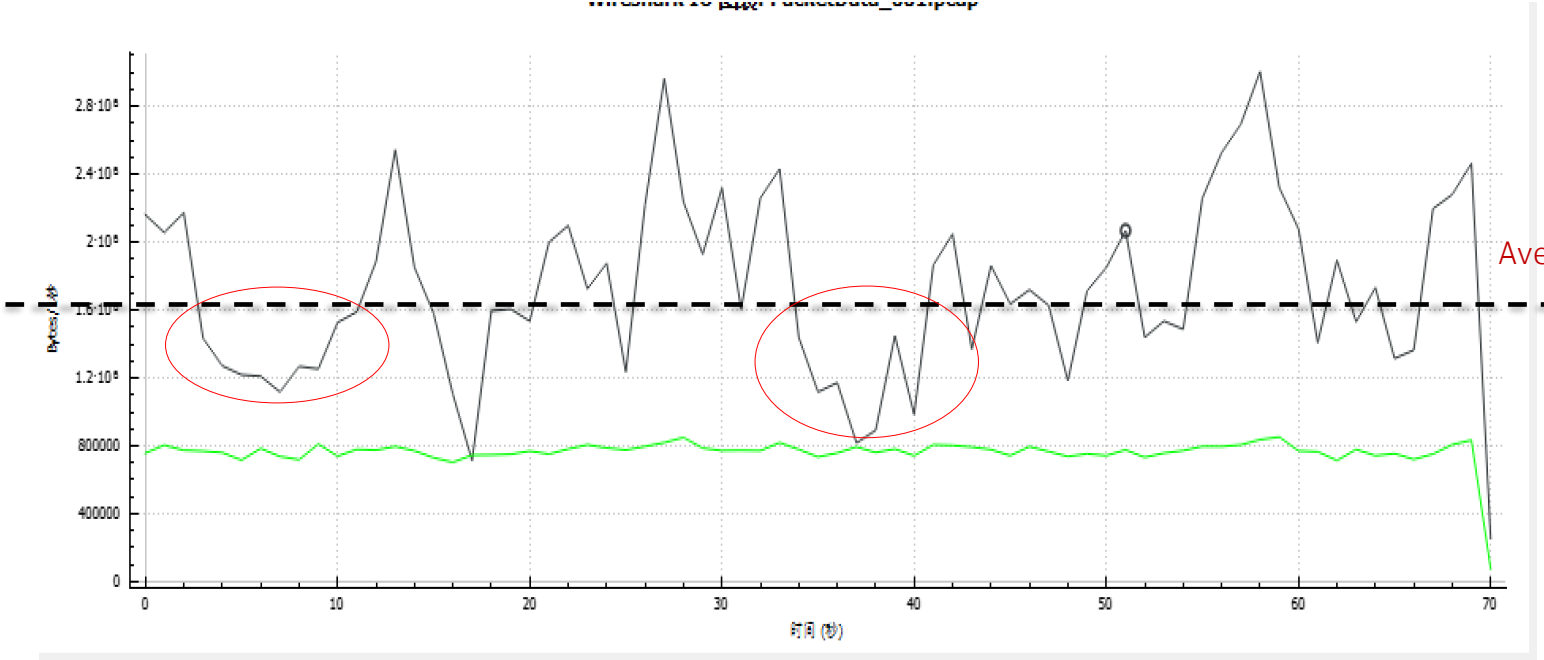


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Issued use case: VNF4 to West SAC DRA

Dallas DLB 04 ← → tsa06.drawsc01.dra.epc.mnc260.mcc310.3gppnetwork.org

Average Sending Speed: 1.6 MB) (800KB ~ 2.8MB)



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Conclusion of above trace

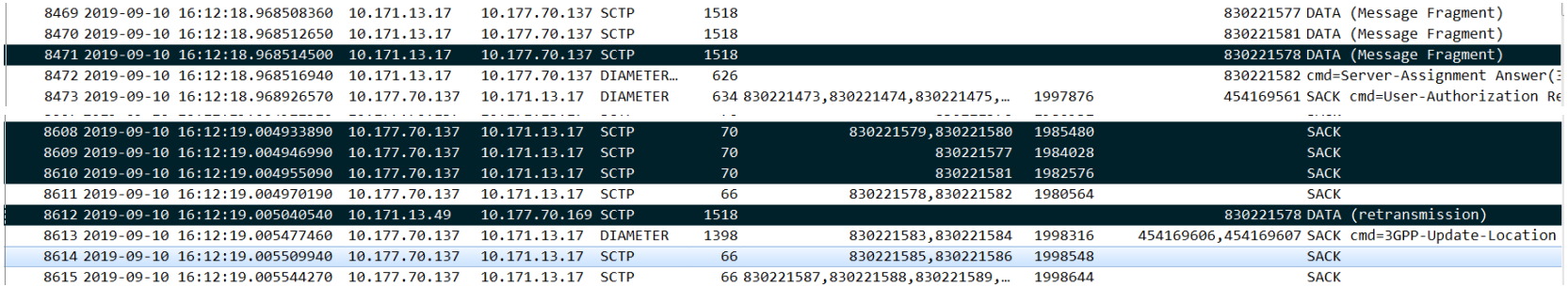
1. Increasing Buffer Size will NOT change the curve (sending rate) which was due to network situation.
2. Increasing Buffer Size can help improving tolerance and reduce problematic connections impact to rest connections.
3. Next Step:
    1. Continue analyze the tuning on # of threads, in relation to the # of connections/associations.

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Early Re-Transmission

1. Refer to section 7.2.4 of RFC4960, the Re-transmission is caused by Gap ACK instead of MinRto(300ms).
2. Re-transmission will be triggered if 3 gap ack for the same TSN.
3. Network jitter make the Re-Transmission looks like race condition, but actually not.

Two example on VNF04.
1. Package 830221578 is sent in frame 8471.
2. In frame 8608, 8609 and 8610 report 830221578 as gap ack.
3. HSS re-transmit 830221578 in 8612.
4. SACK to 830221578 is sent in 8611. This should be sent following 8612, but the order is changed. It should be caused by Network delay or jitter with NetScout.



8469	2019-09-10 16:12:18.968508360	10.171.13.17	10.177.70.137	SCTP	1518				830221577	DATA (Message Fragment)
8470	2019-09-10 16:12:18.968512650	10.171.13.17	10.177.70.137	SCTP	1518				830221581	DATA (Message Fragment)
8471	2019-09-10 16:12:18.968514500	10.171.13.17	10.177.70.137	SCTP	1518				830221578	DATA (Message Fragment)
8472	2019-09-10 16:12:18.968516940	10.171.13.17	10.177.70.137	DIAMETER…	626				830221582	cmd=Server-Assignment Answer(3
8473	2019-09-10 16:12:18.968926570	10.177.70.137	10.171.13.17	DIAMETER	634	830221473,830221474,830221475,…	1997876		454169561	SACK cmd=User-Authorization Re
8608	2019-09-10 16:12:19.004933890	10.177.70.137	10.171.13.17	SCTP	70	830221579,830221580	1985480			SACK
8609	2019-09-10 16:12:19.004946990	10.177.70.137	10.171.13.17	SCTP	70	830221577	1984028			SACK
8610	2019-09-10 16:12:19.004955090	10.177.70.137	10.171.13.17	SCTP	70	830221581	1982576			SACK
8611	2019-09-10 16:12:19.004970190	10.177.70.137	10.171.13.17	SCTP	66	830221578,830221582	1980564			SACK
8612	2019-09-10 16:12:19.005040540	10.171.13.49	10.177.70.169	SCTP	1518				830221578	DATA (retransmission)
8613	2019-09-10 16:12:19.005477460	10.177.70.137	10.171.13.17	DIAMETER	1398	830221583,830221584	1998316	454169606,454169607		SACK cmd=3GPP-Update-Location
8614	2019-09-10 16:12:19.005509940	10.177.70.137	10.171.13.17	SCTP	66	830221585,830221586	1998548			SACK
8615	2019-09-10 16:12:19.005544270	10.177.70.137	10.171.13.17	SCTP	66	830221587,830221588,830221589,…	1998644			SACK

Conclusion of above trace: the re-transmission could be well explained. It's not an issue

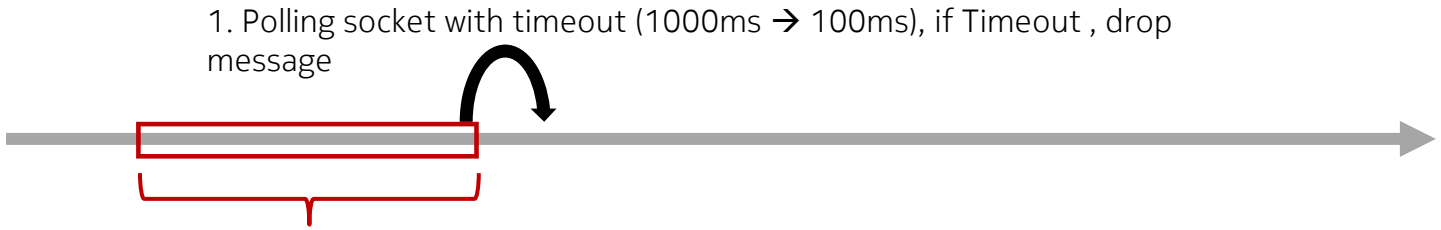
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Eliminate impact to healthy connections

- ✓ **Isolate unhealthy connections**
  - disconnect unhealth connection(Not work as expected, DRA has retry mechanism)
  - increase thread number （4 thread-pairs → 24 thread pairs）
- ✓ **“Non-blocking/Async IO”**
  - Drop messages on blocked connection
  - Buffer message on blocked connection and continue process other message

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Solution to isolate unhealthy connections



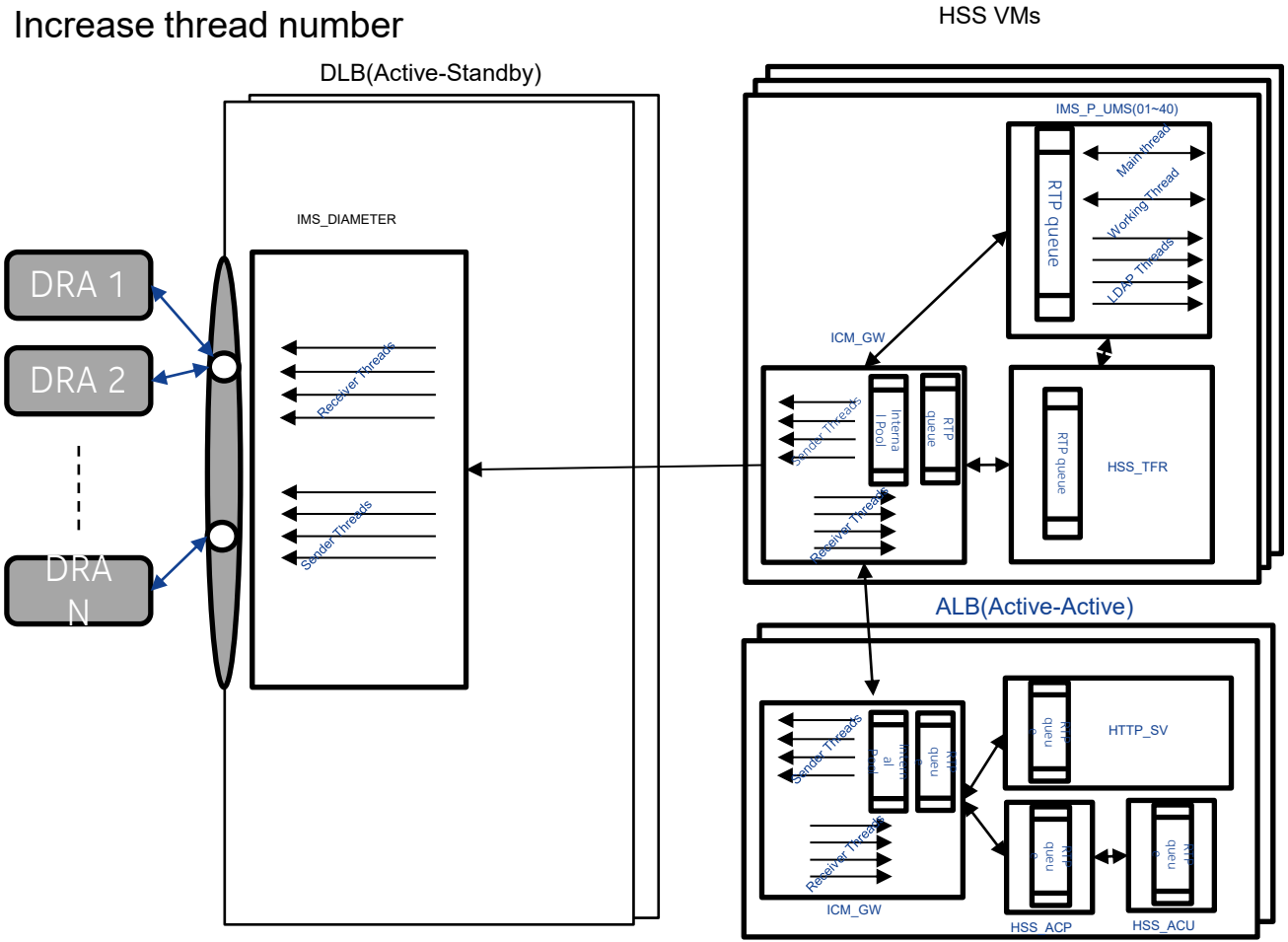
Solution :

1. **When SCTP socket send buffer is full, DLB will try sending/writing packet with timed wait of 100msec**
2. **If fails to write even with 100msec, it will discard the packet and count it as failed attempt**
3. **10 such attempts will be counted in a window of 5sec and SCTP association will be teared down**
4. **This timed wait will force other packets ready to be sent out to go in to wait state adding to delay**
5. **If more such attempts fail, more is the latency experienced by other packets ready to be sent out**

Poll Timeout change to 0, drop message not blocking

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Increase thread number



1. HSS VM CPU/memory
2. RSS Queue number
3. Heap Size Audit
4. ICM connection number tunning
5. IMS_DIAMETE thread number tunning

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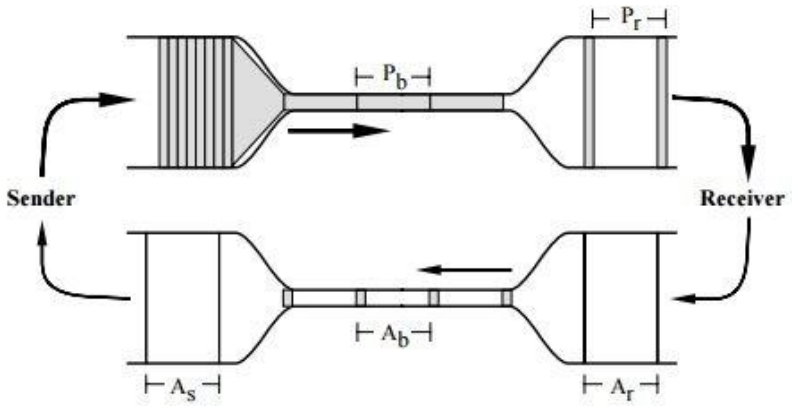
Solve Single connection congestion issue

- ✓ **Increase single connection throughput**
  - better congestion algorithm
  - Parameter tuning with current congestion algorithm
  - Use TCP connection instead of SCTP
  - Increase RSS queue number
- ✓ **Decrease single connection throughput**
  - increase sctp connection number
    - Increase VNF node (DLB number)
    - Increase sctp connection number per VNF node

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Increase single connection throughput

Parameter tuning with current congestion algorithm



The goal is to utilize network resource more efficiently. The limitation caused by window size can be calculated as follows:

$$\text{Throughput} \leq \frac{CWIN}{RTT}$$

where CWIN is the Congestion Window and RTT is the round-trip time for the path.
RWIN <= SCTP buffer Size /2

NO.	Buffer Size	WINS	RTT	single connection throughput	TPS(2K/message)
1	200K	100K	100ms	1 M/s	500TPS
2	2M	1M	100ms	10M/s	5000TPS

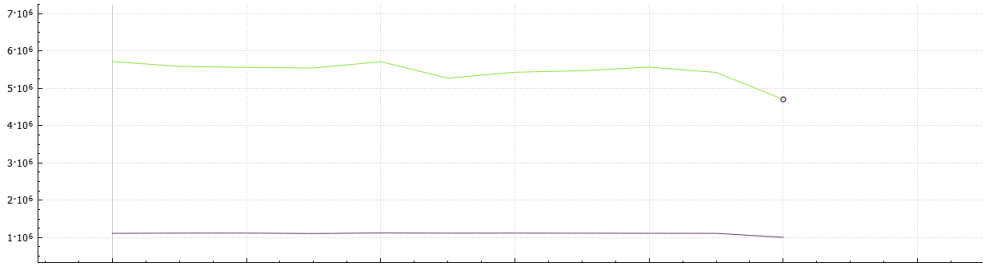
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Test Result And Summary – RSS:8, Sending Buffer:2M ---BM PT

Scenario	Total TPS	TPS/Connection	Mbps	Delay	Jitter	Avg Message Size	Packet Loss	Send Buffer MB/Conn	Receive Buffer on Test Client	Dia Connection	No. of Test Client	RSS Queue	Threads	SCTP Buffer Usage	# Connection Resets	Message response time(ms)
BM PT	456	456	7.2	140ms	0	1k	0	200K	4M	1	1	8	4	115K	no	avg:151,min:144,max:180
BM PT	6764	6764	54	140ms	0	1k	0	2M	4M	1	1	8		1622K	no	avg:144,min:143,max:149
BM PT	4370	4370	34.97	140ms	0	1k	0	2M	4M	1	1	8		1045k	no	avg:152,min:143,max:209

Did the similar testing on BM to compare the difference.

1. BM PT support 456 TPS with 200K sending buffer size with 140ms Wan-Delay
2. >10 times traffic increased to 6764 tps after increasing 200K->2M sending buffer. Need to increase BM T
3. No any packet loss and the traffic is very stable.



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Test Result And Summary – RSS:3, Sending Buffer:200K, 2M

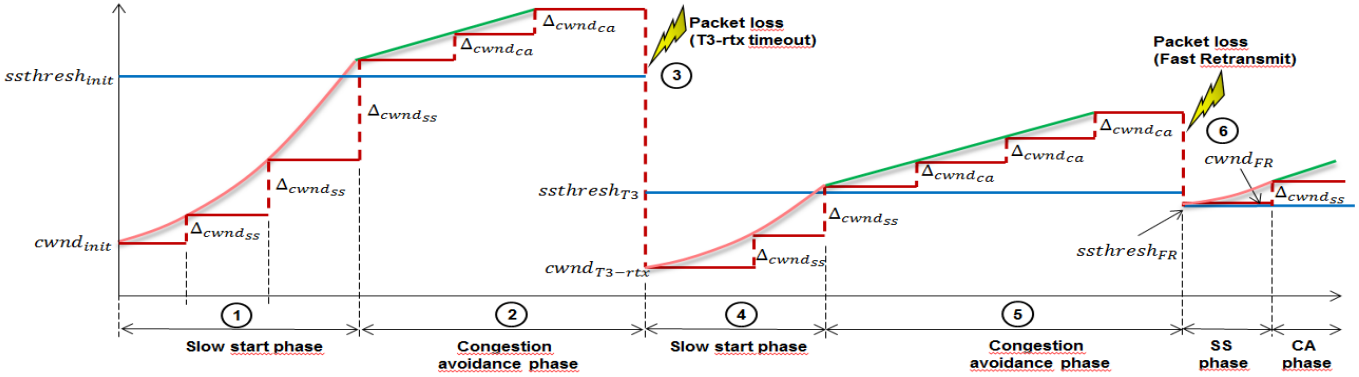
Lab Type	RSS Queue	Buffer Size	Wan-Delay	No.of Link	PollTimeoutOnSocketSendError	EAGAINMonitoringMaxCount	EAGAINMonitoringInterval
VNF HSS	1	2M	140ms	1-6	0ms	20	5s
VNF HSS	3	200K	140ms	1-6	0ms	20	5s
VNF HSS	3	2M	140ms	1-6	0ms	20	5s
VNF HSS	8	2M	140ms	1-6	0ms	20	5s
BM	8	200K	140ms	1-6	0ms	20	5s
BM	8	2M	140ms	1-6	0ms	20	5s

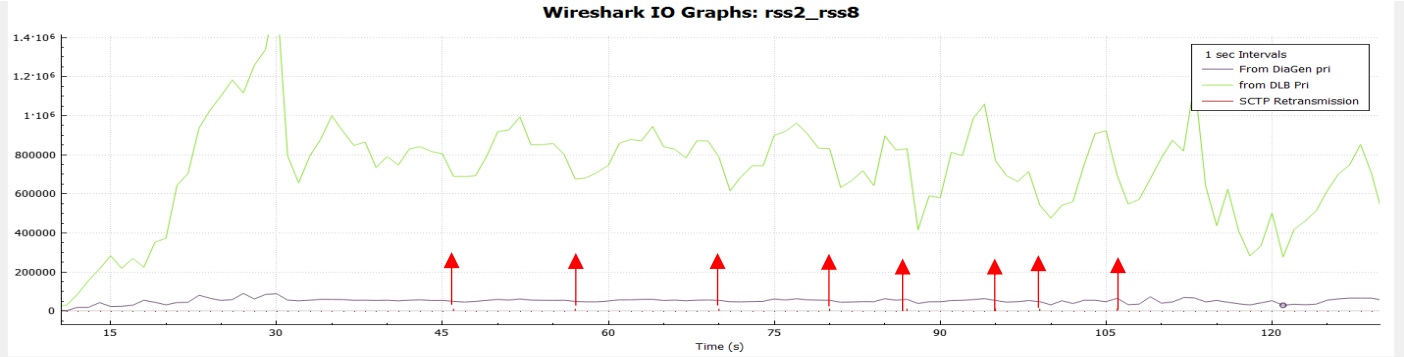
Test based on above configuration in green.
Avg message size:~1k
Observation:
TPS is 380TPS after enabling RSS = 3 with 200K buffer (TMO configuration after 1st patch).
TPS is 950TPS after enabling RSS = 3 with 2M buffer.

Scenario	Total TPS	TPS/Connection	Mbps	Delay	Jitter	Avg Message Size	Packet Loss	Send Buffer MB/Conn	Receive Buffer on Test Client	Dia Connection	No. of Test Client	RSS Queue	Threads	SCTP Buffer Usage	# Connection Resets	Comments
VNF	11200	11200	100	20ms	0	1k	0	2M	4M	1	1	1	4	<500K	No	The limitation is not on DLB and sending buffer. It's Callp and test client's limitation to reach higher TPS.
VNF	9500	9500	76	140ms	0	1k	0	2M	4M	1	1	1	4	~2M	No	The limitation is not on DLB and sending buffer. It's Callp and test client's limitation to reach higher TPS.
VNF	950	950	7.6	140ms	0	1k	0	2M	4M	1	1	3	4	300K~2.5M	No	Traffic stable
VNF	380	380	3.04	140ms	0	1k	0	200K	4M	1	1	3	4	400K~290M	NO	Traffic stable

Reference: SCTP Congestion Control

RFC 4960: 3 missing reports -> Loss -> Fast Retransmit





One real example on lab the red arrow is SCTP.Retransmission

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Testing Result Analysis – Why Out-Of-Order matters

1. Out-of-order chunks triggers gap SACK with missing TSN report, example:
   - Correct TSN sequence: 1, 2, 3, 4, 5, 6, 7, 8
   - Actual TSN sequence delivered to receiver: 1, 5, 2, 6, 3, 7, 4, 8
   - When receiving 5, receiver reports 2, 3, 4 as missing; 3, 4 reported as missing upon 6; 4 reported as missing upon 7
   - So 4 is reported as missing for 3 times
2. A total of 3 missing (loss) reports triggers Fast Retransmit

https://tools.ietf.org/html/rfc4960#section-7.2.4

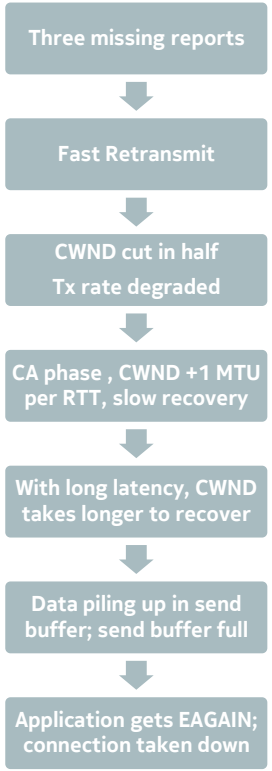
3. Fast Retransmit causes CWND cut in half, and eventually into Congestion Avoidance phase

https://tools.ietf.org/html/rfc4960#section-7.2.3

4. In Congestion Avoidance phase, CWND is incremented by only 1*MTU per RTT

https://tools.ietf.org/html/rfc4960#section-7.2.2

5. With longer latency (RTT), CWND needs much longer to recover to that before cut
6. Before CWND recovers, sending buffer is piling up, as less data can be sent out (smaller CWND)
7. Application gets EAGAIN when send buffer is full; with several EAGAIN, connection is taken down



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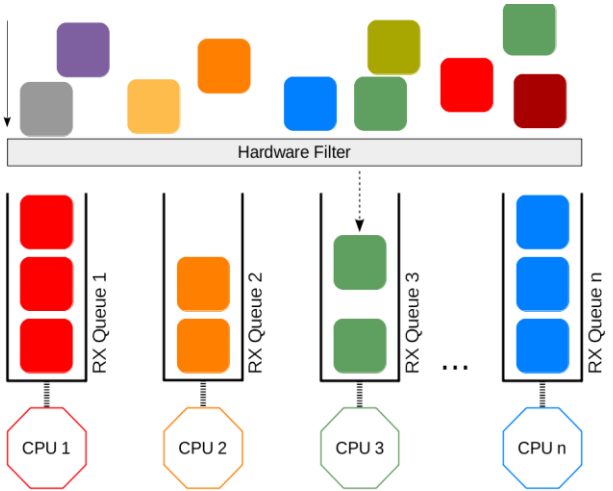
Technical Analysis – Receiving Side Scaling (RSS) in Bare Metal

Receiving Side Scaling (RSS):

- Physical NIC with multiple Rx queues (optionally, multiple Tx queue)
- Filter (usually hash function) can be invoked on every Rx packet (SA, DA, SP, DP), to get a hash value
- Same hash value is derived from all packets of the same connection
- With same hash value, all packets of the same connection go the same Rx queue
- As in same Rx queue, Rx packets of same connection are processed in order
- Tx queue selected based on Rx queue, or CPU, depending on NIC driver (ndo_select_queue); Tx is in order
- No Out-Of-Order observed

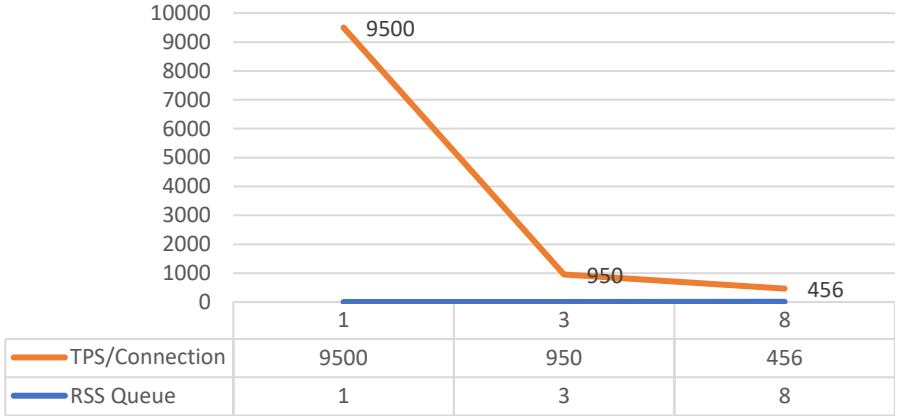
Testing proved on BM HSS

- 8 RSS queue
- With RTT 140ms, 2M sending buffer
- Single connection can still reach high throughput

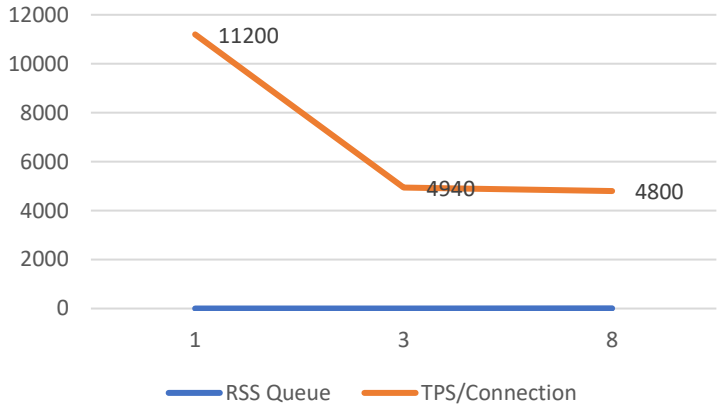


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Increase RSS queue number (TMO has increased RSS queue number)



140ms Wan-Delay, No Packet Loss.



20ms Wan-Delay, No Packet Loss.

Observation:

1. Many disordered packet sent out from DLB when multiple queue (RSS) is enabled.
2. Disordered packet cause SCTP fast re-transmission and single link capacity dropped dramatically.

Proposal To Improve the VNF Link Capacity:

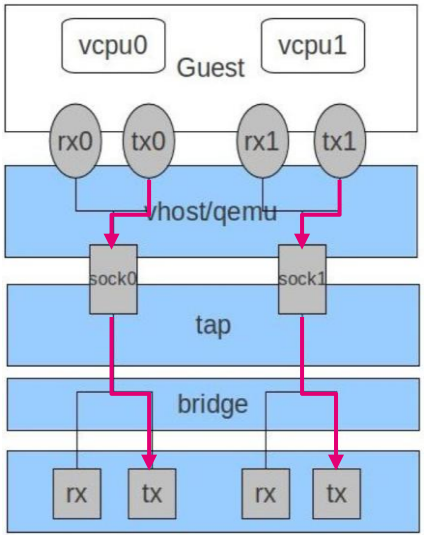
1. Enable single queue to validate 80KTPS traffic
2. Resolve the packet disorder issue when multiple queue is enabled – long term solution

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Final Root Cause

- Multiqueue (virtio-net + vhost-net) in VNF is not real RSS:
  - A lot of features missing (get/set hash, statistics, etc.)
  - Always symmetric number. of Rx queue & Tx queue
  - As of CentOS7.6, Tx queue is selected based on CPU, not XPS, not hash
    - virtio-net driver in Guest has taken over the control by setting ndo_select_queue
    - So tx queue selection of __netdev_pick_tx is bypassed, including XPS, checking of ooo_okay
    - There is a later change of virtio-net to restore the control, however, that is not in CentOS7.6
  - Unlike RSS, in KVM Multiqueue, Rx queue to Guest is determined by Tx queue
    - So Rx queue switch is also possible due to Tx queue switch (which is due to different CPU)

```
struct netdev_queue *netdev_pick_tx(struct net_device *dev, struct sk_buff *skb, void *accel_pri)
{
    if(if (ops->ndo_select_queue))
    {
        ndo_select_queue()
    }
    else
    {
        __netdev_pick_tx
        {
            //get XPS queue ,OOO checking
        }
    }
}
```



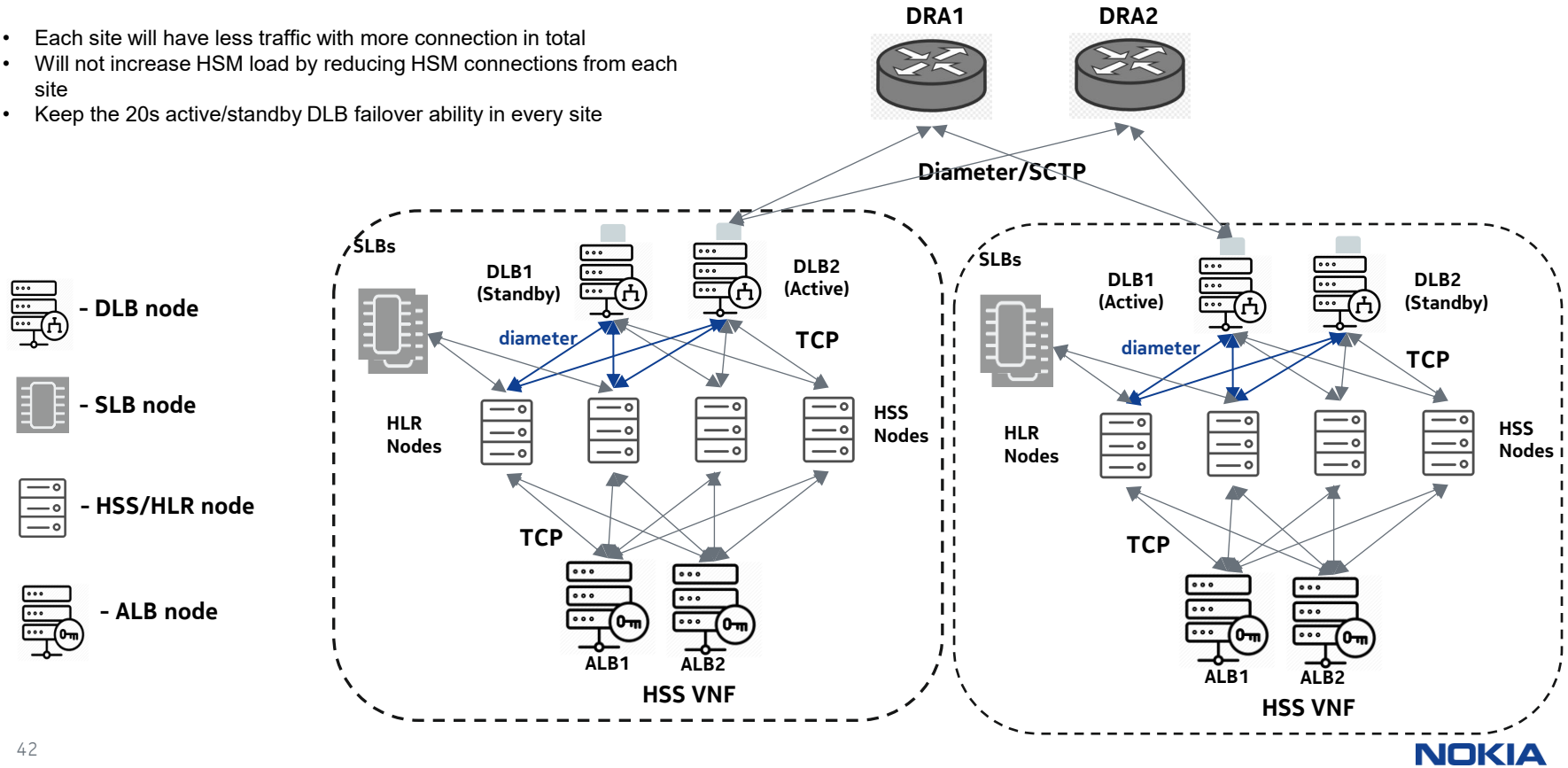
Kernel driver patch: https://lore.kernel.org/patchwork/patch/409812/

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Decrease single connection throughput

--Split one VNF to Multi VNFs

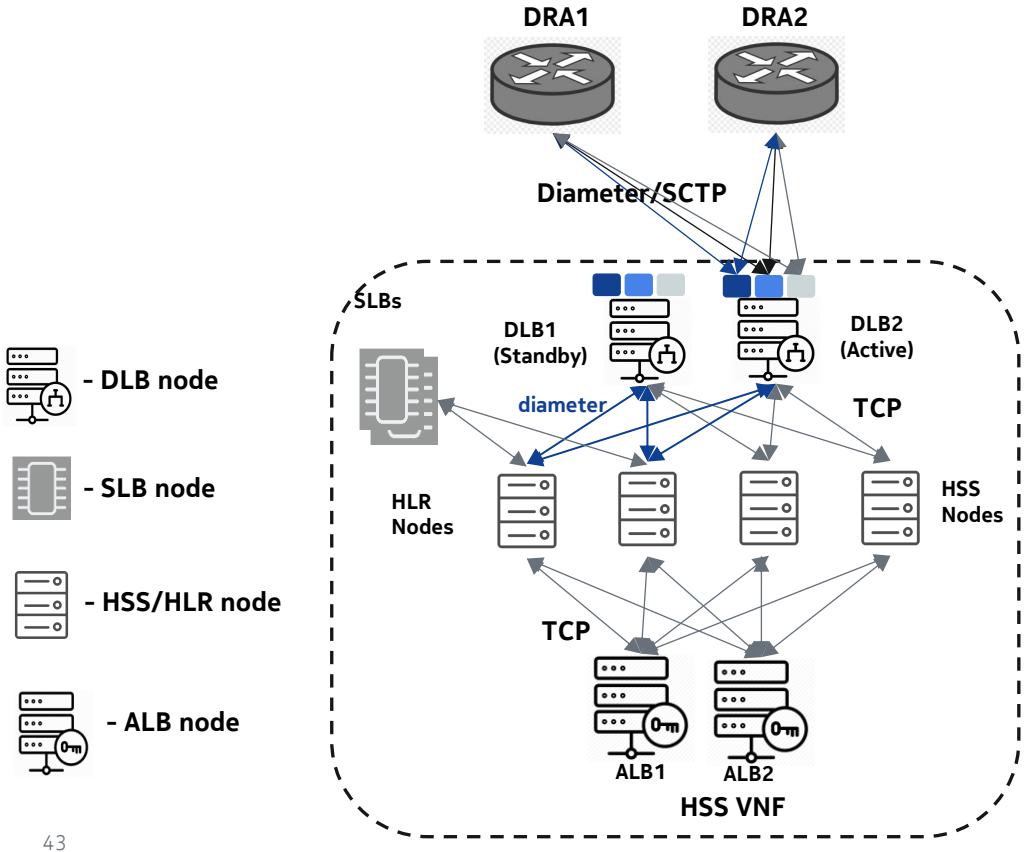
- Each site will have less traffic with more connection in total
- Will not increase HSM load by reducing HSM connections from each site
- Keep the 20s active/standby DLB failover ability in every site



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Decrease single connection throughput

--increase connection per vm



✓ **DRA initiate more connections to HSS**

- DRA must support configure multiple connection to one HSS
- No changes from HSS side

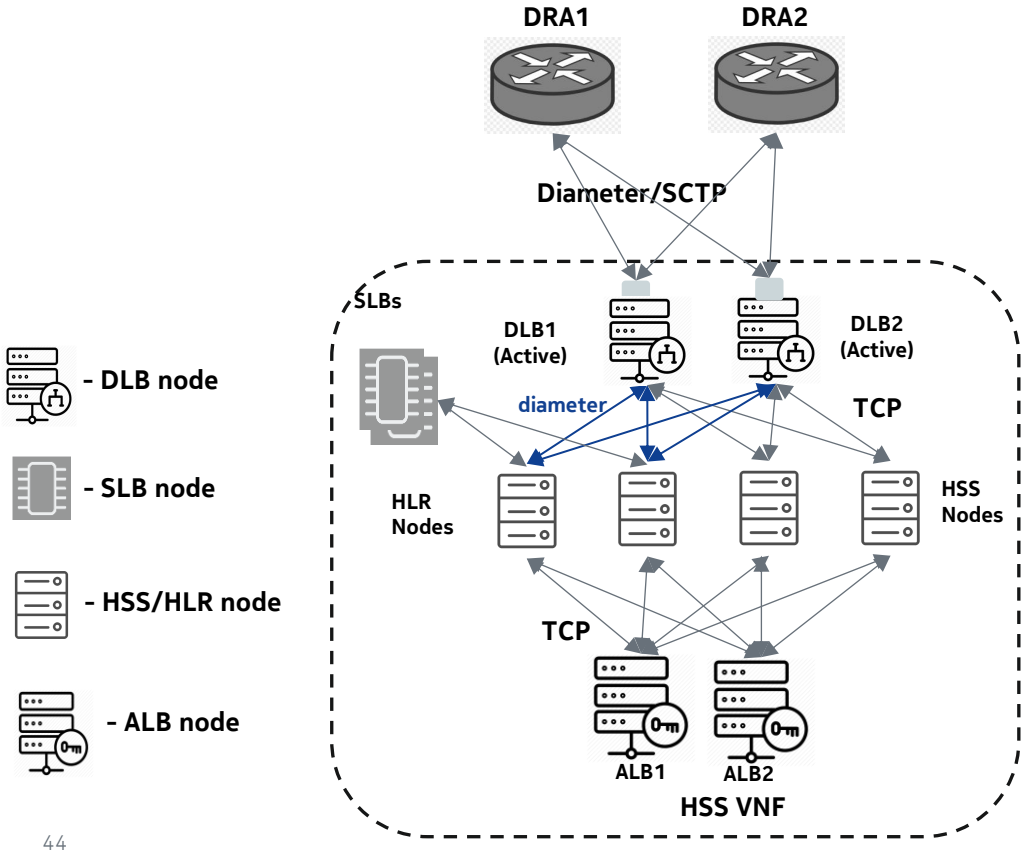
✓ **HSS listening on more ports**

- Use multiple ports to listening connection: 3868, 3869 (or 3870, etc.)
- The usage of 3869 (or 3870, etc.) is not standard. Need DRA support.
- Already used by Other customer(per interfaces)

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Decrease single connection throughput

--Active-Active DLB



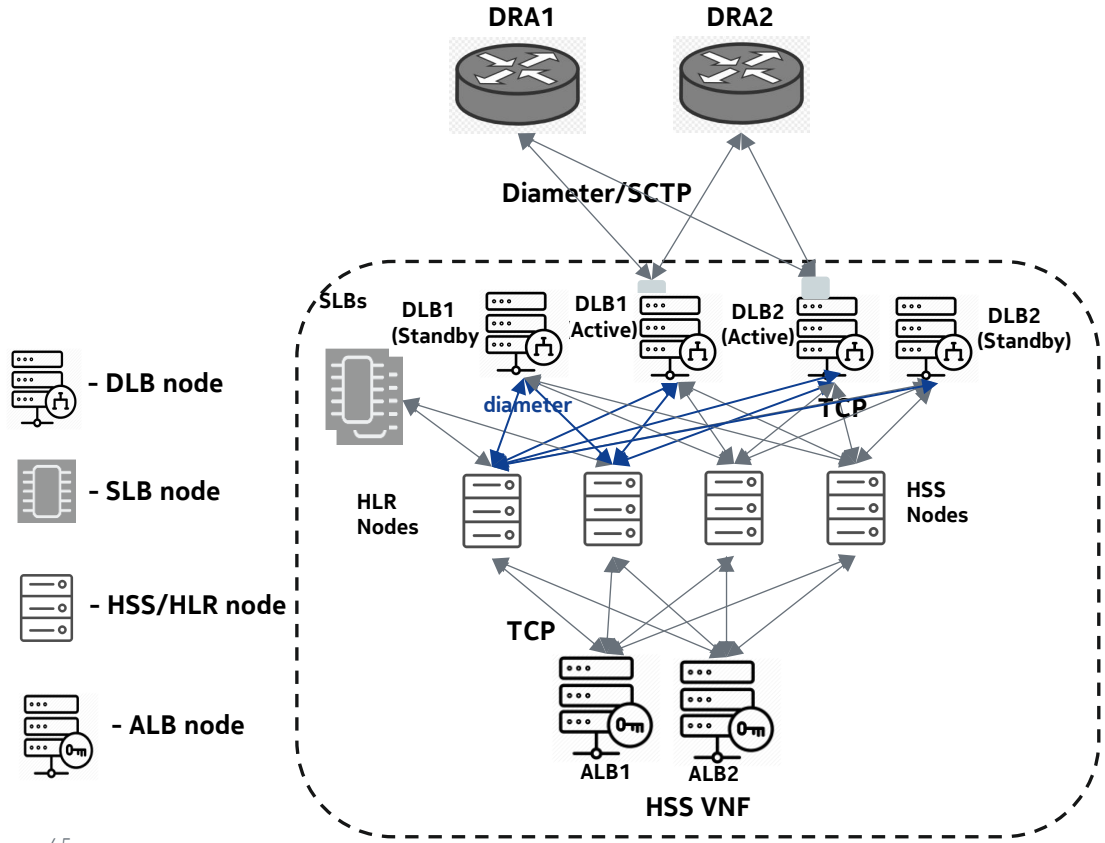
✓ **Change Active-Standby DLB to Active-Active**

- Need ICM to support Active-Active mode
- Floating IP s not needed

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Decrease single connection throughput

--Add one pair A/S DLB



✓ **Change Active-Standby DLB to A/S-A/S**

- Need ICM to support Active-Active mode
- Floating IP s not needed

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Decrease single connection throughput --Other Proposals

- One pair of A/S DLB supporting a number of Callp

  Proposed by customer
  Our suggestion is no-go because:

  1. this will result in multiple clusters and it breaks Platform's fundamental design of inter-VM communication
  2. even if Platform could do this, HSS will also significant code changes
  3. huge regression, stability and performance testing will also be needed due to this change.

- A-A-A

  DLB scaling Active-Active-Active Architecture

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Other Actions

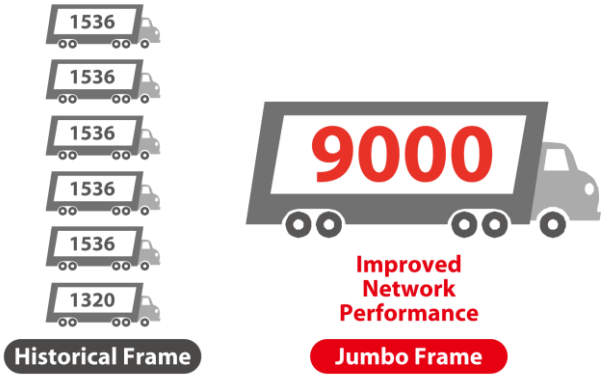
- SCTP Buffer Monitor

  Monitor all SCTP association SCTP buffer Usage, raise alarm if threshold reaches
- Poll → epoll / Architecture change (Wally)
- SCTP Performance Test(Jumbo Frame MTP from 1.5K to 9K)
- A-A DLB enhancement
- etc

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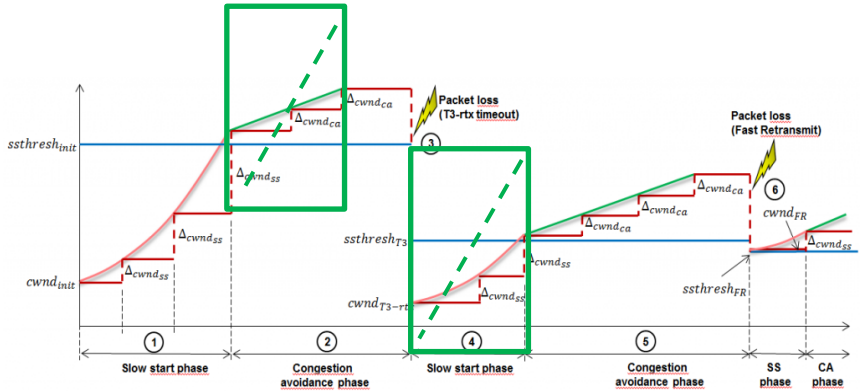
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Better Performance with Jumbo Frame



1. **Less frame number means less retransmission**

2. **Bigger MTU Size means faster recovery speed**



Note: this is still under test, no comprehensive test result yes

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Some thoughts for delay issues

1. When this fault should be found ?

- During Architecture design, FRS should have clear description about single connection throughput and minimum connection numbers and other resources(CPU/memory/Disk/Network bandwidth) required to meet overall performance (This should be defined for other interface: ldap, UHHD,soap, HSM etc) .
- PT Test and dimensioning should first follow FRS requirement and take WAN delay, Peer Node's configuration etc into consideration

2. How to debug if this kind of issue happens?

- **Setup PT Lab to reproduce issue(Issue reproduction plays a decisive role in both outage)**
- Identify the bottleneck with logs/trace (Wireshark trace of both diameter and icm)
- Need better knowledge of OS/network stack /Wireshark tool to analyze log/trace.

3. What needs to be enhanced ?

- How to design Log/Alarm/Counter system for distributed system When issue happened, current log/alarm/counter are not helpful.
- How to Design for Software Robustness and Resiliency in High-Performance and distributed environment (Singe point of failure/bottleneck, multiple-threads Async IO ,overload system)
- Performance Tuning skill (network connection, CPU, Memory Usage etc)

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